Chapter 6

Free Electron Theory of Metals

We know that more than a two-third of elements are metals. This shows that the natural tendency of elements is to favour the metallic state. Of the common solids most of which are insulators, metals have found the largest number of applications according them thereby a very special status. The striking physical properties that make them so attractive should emerge as natural consequences of a successful theory. It is easy to develop a theory for metals because of their relatively simple structure. Then, this theory can be extended to other solids with suitable modifications as the situation demands. As no perfect theory exists even today, we make an endeavour in this chapter to give an appropriate account of the main theoretical models proposed in order to explain the characteristic properties of metals.

The theory of metals heavily relies on the conviction that the valence electrons of metal atoms can be treated as free. The valence electrons are so loosely bound to the ion core that in the first approximation they may be considered to have freedom of moving through the volume of the metal. For example, the so called free nature of these electrons readily explains the metallic lustre as for free particles the emission of radiation immediately follows the process of absorption. The concept of free electron motion is indicated by the observed large electrical and thermal conductivities of metals. The free electrons in metals are called '*conduction electrons*' since they are responsible for the electric conduction. The collective body of conduction electrons contributed by individual metal atoms is conventionally interpreted as '*the free electron gas*' that occupies most of the volume of the metal. The fraction of volume belonging to the positive metal ions is mere 15 per cent in sodium metal. To sum up, a metallic crystal may be envisaged as the superposition of a periodic array of positive ion cores and a quasi-uniform negative charge density of the free electron gas.

The formulation of the theory of metals began well before the announcement of the Pauli principle (1926) and the advent of quantum mechanics (1925). Drude was the first to propose a model in 1900, three years after the discovery of electron by J.J. Thomson. Though the model operates in the domain of classical physics, its importance cannot be undermined in view of the amazing success it has achieved in deriving the Ohm's law and the relationship between the electrical and thermal conductivities. Its another feature of quickly demonstrating the behaviour of electrical conductivity further emphasizes the value of its contribution. A befitting account of this theory will be given in this chapter with a final word on the classical models. With the Pauli exclusion principle coming to force, the electrons in metals could no longer be treated by the classical statistics. The requirement that the electrons be subjected to the Pauli principle brought about a remarkable transformation in the theory which is now capable of addressing most of the questions left unanswered by the classical theory. This replaced the Maxwell–Boltzmann statistics by the Fermi–Dirac statistics and renamed the free electron gas as '*the free electron Fermi gas*'. Sommerfeld (1927) won the distinction of applying the related quantum mechanical ideas for the first time to explain the properties of metals. A suitable account of the Sommerfeld theory will lead us to a concluding discussion of this chapter.

6.1 THE DRUDE MODEL

The Drude model is essentially based on the classical kinetic theory of gases. According to Drude the metal must have two types of particles as against only one type in the simplest gases. The discovery of electrons bearing negative charge made it mandatory to accept the presence of positively charged entities (or particles) on the requirement of the charge neutrality condition. It is assumed that when metal atoms come together to form a metal, the valence electrons get liberated and move freely within the metal. The remainder of the atom is a positive ion carrying the major portion of the atomic mass. Drude took these particles as heavy and immobile. Under the action of an external electric field the free electrons referred to as conduction electrons in metals move in the background of immobile positive metal ions. In the absence of a relevant theoretical framework to deal with the free electron gas, Drude took recourse to the methods of kinetic theory of ideal gases for examining metallic conduction without getting deterred by the large electron densities (10^{22} cm^{-3}). It is assumed that the reader is acquainted with the postulates of kinetic theory of gases through a prior exposure, and therefore assumptions specific to only free electron gas are being given below:

1. The electron–electron and electron–ion interactions are neglected. The approach tantamounts to working in the independent electron approximation and the free electron approximation in that order. To be exact, the free electron approximation is not adhered to in strict literal sense because the electrons are considered to remain confined within the metal in the Drude Model and this is possible only if the electron–ion attractive force is active.
2. Under the action of an external electric field, electrons move opposite to the field direction and make collisions with immobile and impenetrable ion cores. An electron bumps† from ion to ion and between successive collisions moves in a straight line as determined by the Newton's equations of motion.
3. All electrons move with the RMS speed of a Maxwell–Boltzmann distribution, representing the random or thermal velocity of electrons. The average electron velocity immediately after the collision is zero.

6.1.1 Electrical Conductivity of Metals

DC Conductivity

We apply Drude theory first to derive an expression for dc conductivity. An external static electric field can affect electron velocity during the time interval between two successive collisions. But the gain in velocity is destroyed each time a collision occurs since the average velocity immediately after the collision is zero. A large influence of electric field is reflected in a larger mean free time or relaxation time τ. The probability of collision per unit time is defined as $1/\tau$. Therefore, the probability that a collision occurs in a small time interval dt is simply dt/τ.

Taking the acceleration of an electron of mass m as eE/m, the mean drift velocity can be written as

$$\mathbf{v}_d = \left(- \frac{e\mathbf{E}}{m} \right) \tau \tag{6.1}$$

† The collision involving physical contact is unrealistic. We, however, refrain from identifying the true scattering mechanism at this stage.

If there are a total of n electrons per m^3 in the metal, all with constant drift velocity $\mathbf{v}_d$, we have the following relation for the net electric current density $\mathbf{j}$:

$$\mathbf{j} = -ne\mathbf{v}_d = \left(\frac{ne^2\tau}{m}\right)\mathbf{E}$$

$$= \sigma\mathbf{E} \tag{6.2}$$

where the positive scalar quantity

$$\sigma = \frac{ne^2\tau}{m} \tag{6.3}$$

is defined as the electrical conductivity (the reciprocal of the electrical resistivity ρ). The form of the expression (6.3) remains the same in all models including those based on quantum physics. The difference lies only in the way n, m, and τ are defined.

We are now beginning to accomplish our aim of highlighting the gains of the Drude model. The first and the foremost of these is the derivation of the Ohm's law contained in (6.2). The law was initially set up purely on an empirical basis.

It must be, however, stated that the electrical conductivity σ is not universally scalar since in some complicated situations $\mathbf{j}$ becomes nonlinear with respect to $\mathbf{E}$ making σ to behave as a tensor. The electrical conductivity is often expressed in terms of the drift mobility of electrons μ as

$$\sigma = ne\mu \tag{6.4}$$

where

$$\mu = \frac{|\mathbf{v}_d|}{\mathrm{E}} \tag{6.5}$$

Drude put (6.3) in a further useful form by exploiting the ideas of kinetic theory such as to define τ in terms of thermal velocity v_{th} expressed by the following relations:

$$\tau = \frac{\Lambda}{v_{th}} \tag{6.6}$$

$$\frac{3}{2}k_BT = \frac{1}{2}mv_{rms}^2 = \frac{1}{2}mv_{th}^2 \tag{6.7}$$

where Λ is the electron mean free path. Using these relations, (6.3) can be written as

$$\sigma = \frac{ne^2\Lambda}{(3mk_BT)^{1/2}} \tag{6.8}$$

The relation (6.8) is another achievement of the Drude theory since it gives the right magnitude of electrical conductivity and correctly describes that conductivity increases with decrease in temperature (see Fig. 6.1). Experimental values at 77, 273 and 373 K are given in Table 6.1 for comparison. But the dependence on $T^{-1/2}$ is not in agreement with the observed T^{-1} dependence in

Table 6.1 Experimental electrical conductivities* of some metals (in 10^8 ohm^{-1} cm^{-1})

Metal	*At* 77 K	*At* 273 K	*At* 373 K
Li	0.96	0.12	0.08
Na	1.25	0.24	melts
K	0.72	0.16	melts
Cu	5.00	0.64	0.45
Ag	3.33	0.66	0.47
Au	2.00	0.49	0.35
Al	3.33	0.41	0.28
Fe	1.52	0.11	0.07
Zn	0.91	0.18	0.13
In	0.56	0.12	0.08
T1	0.27	0.07	0.04
Pb	0.21	0.05	0.037

*From G.W.C. Kaye and T.H. Laby, *Table of Physical and Chemical Constants* (Longmans Green, 1966).

most of the common metals. The modern theory that resolves this issue attributes this inconsistency to the unrealistic assumption of electron–ion elastic collisions and the use of classical statistics to describe the electrons in the Drude model. The Drude model is further plagued by its inability to account for the low electron heat capacity and the temperature independent behaviour of paramagnetic susceptibility of conduction electrons. As we will see later in this chapter and in Chapter 13, solutions to both of these problems are achieved by applying quantum statistics.

By feeding the measured value of conductivity in (6.3), we obtain the value of relaxation time τ. The estimated value of τ is in the range 10^{-15} to 10^{-14} s. The thermal velocities estimated from (6.7) are around 10^5 m s^{-1}. Using (6.6), we find that the typical values of mean free path Λ are of the order of a few angstroms which compare with the interatomic separations. This in a way supports the electron–ion collisions in Drude theory. But the low temperature measurements on some of the purest and least imperfect crystals show the mean free path as large as a few centimetres. This strongly contradicts the Drude picture in which the electron bumps along from ion to ion. On the other hand, the modern theory tells that electron waves cannot be scattered by a perfectly periodic potential leading to an infinite mean free path in an ideal crystal. The experimental data in no way casts aspersion on the above theoretical view, simply because no body has ever produced any perfect and ideal crystal. Since every real crystal has deviations from periodicity because of the presence of imperfections and even impurities, the value of the mean free path is limited by the scattering from these centres. The measured values of electrical conductivity of a highly pure sample of copper metal is plotted as a function of temperature in Fig. 6.1.

AC conductivity

Here, we need to calculate the current that would flow under the influence of an alternating electric field (time-dependent),

$$\mathbf{E} = \mathbf{E}(\omega)e^{-i\omega t} \tag{6.9}$$

If the time-dependent momentum per electron be denoted by $\mathbf{p}$, we get the following equation of motion for the electron,

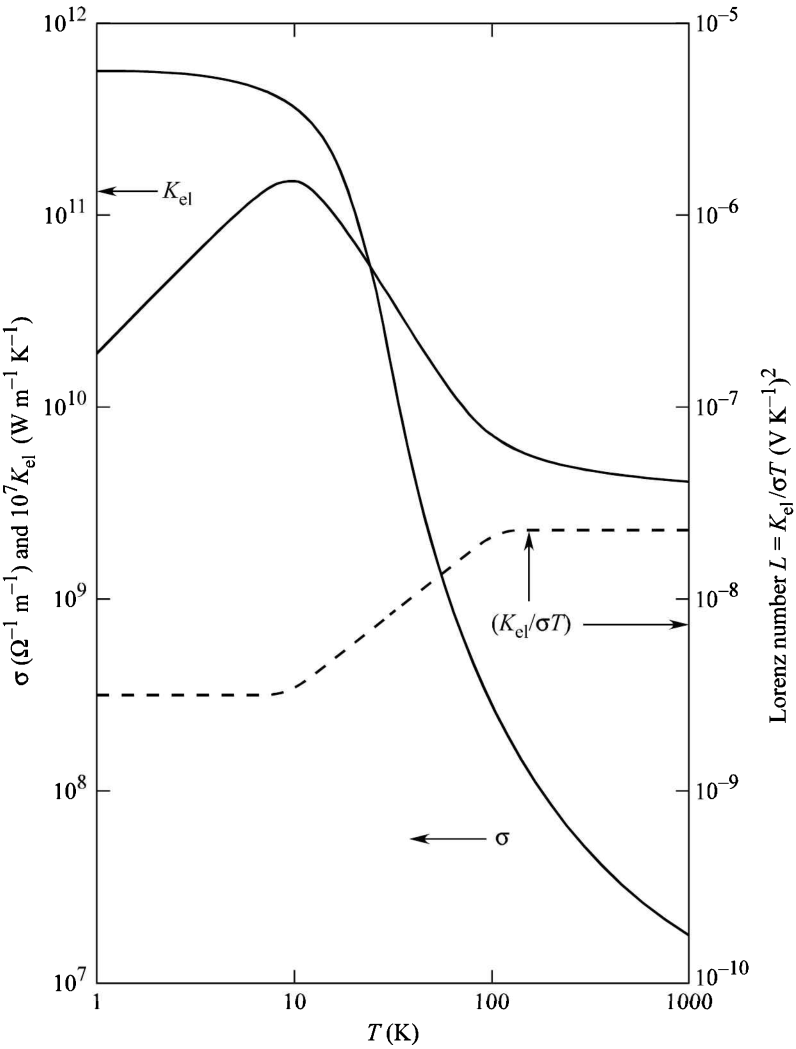

FIG. 6.1 Variations in electrical conductivity σ, electronic thermal conductivity K_{el} and Lorenz number L ($= K_{el}/\sigma T$) with change in temperature for a highly pure sample of copper metal. (Data on σ from G.T. Meaden, *Electrical Resistance of Metals* (Plenum Press, 1965); and K_{el} values from *American Institute of Physics Handbook* (McGraw-Hill, 1971).

$$\frac{d\mathbf{p}}{dt} = -\frac{\mathbf{p}}{\tau} - e\mathbf{E} \tag{6.10}$$

where $\left(-\dfrac{\mathbf{p}}{\tau}\right)$ represents the damping term following from collisions.

The time-dependent solution of the equation of motion is sought in the form

$$\mathbf{p} = \mathbf{p}(\omega)e^{-i\omega t} \tag{6.11}$$

On substituting **p** and **E** in (6.10), we get

$$\mathbf{p}(\omega) = -\frac{e\mathbf{E}(\omega)}{\left(\dfrac{1}{\tau} - i\omega\right)} \tag{6.12}$$

Since the current density will also depend on time, it can be written as

$$\mathbf{j} = \mathbf{j}(\omega)e^{-i\omega t} \tag{6.13}$$

But
$$\mathbf{j} = -ne\mathbf{v}_d = -\frac{ne\mathbf{p}}{m}$$

Therefore, using (6.11) and (6.13), we get

$$\mathbf{j}(\omega) = -\frac{ne\mathbf{p}(\omega)}{m}$$

$$= \frac{ne^2}{m\left(\dfrac{1}{\tau} - i\omega\right)}\mathbf{E}(\omega) \quad \text{(using 6.12)} \tag{6.14}$$

One usually writes (6.14) in the form of Ohm's law:

$$\mathbf{j}(\omega) = \sigma(\omega)\mathbf{E}(\omega) \tag{6.15}$$

where
$$\sigma(\omega) = \frac{ne^2}{m\left(\dfrac{1}{\tau} - i\omega\right)} = \frac{\sigma_0}{(1 - i\omega\tau)} \tag{6.16}$$

$\sigma(\omega)$ is called the ac conductivity and σ_0 denotes the dc conductivity (6.3).

It is instructive to find that $\sigma(\omega)$ becomes equal to σ_0 when the ac field is replaced with a dc field (i.e. $\omega = 0$).

Magnetoconductivity

Magnetoconductivity is the term used to denote the electrical conductivity as modified by the application of a steady magnetic field. On grounds of simplicity, we discuss the case of a metal whose conductivity is interpreted in terms of the nearly free electron motion and examine the same in the classical relaxation time approximation.

The Lorentz force controls the motion of nearly free electrons described by the usual equation of motion:

$$m\frac{d^2\mathbf{r}}{dt^2} + \frac{m}{\tau}\frac{d\mathbf{r}}{dt} = -e\left(\mathbf{E} + \frac{d\mathbf{r}}{dt} \times \mathbf{B}\right) \tag{6.17}$$

The second term on LHS represents the damping force that follows from the collisions with an average relaxation time τ. When the magnetic field is impressed along z-direction ($\mathbf{B} = B\hat{\mathbf{z}}$), the components of electron motion along the three Cartesian axes (x, y, z) are given by the following equations:

$$
\begin{aligned}
m\ddot{\mathbf{x}} + \frac{m}{\tau}\dot{\mathbf{x}} &= -e\mathbf{E}_x - m\omega_c\dot{\mathbf{y}} \\
m\ddot{\mathbf{y}} + \frac{m}{\tau}\dot{\mathbf{y}} &= -e\mathbf{E}_y + m\omega_c\dot{\mathbf{x}} \\
m\ddot{\mathbf{z}} + \frac{m}{\tau}\dot{\mathbf{z}} &= -e\mathbf{E}_z
\end{aligned}
\tag{6.18}
$$

where ω_c stands for the cyclotron frequency $\left(\dfrac{eB}{m}\right)$.

We begin with the analysis of the motion along z-direction described by the last of the above equations simply because it is the simplest of the three. For the sake of generality, let us consider the application of an oscillatory electric field,

$$\mathbf{E}(t) = \mathbf{E}(\omega)e^{i\omega t} \tag{6.19}$$

assuming a similar time dependence for the electron instantaneous position ($\sim e^{i\omega t}$).

With these substitutions, the z-component of motion is given by

$$m\left(i\omega + \frac{1}{\tau}\right)\dot{\mathbf{z}} = -e\mathbf{E}_z$$

or

$$\dot{\mathbf{z}} = -\frac{e\mathbf{E}_z}{m\left(i\omega + \dfrac{1}{\tau}\right)} \tag{6.20}$$

It leads to the following expression for the current density,

$$\mathbf{j}_z(t) = -ne\dot{\mathbf{z}} = \frac{ne^2}{m\left(i\omega + \dfrac{1}{\tau}\right)}\mathbf{E}_z(t)$$

or

$$\mathbf{j}_z(\omega) = \frac{ne^2}{m\left(i\omega + \dfrac{1}{\tau}\right)}\mathbf{E}_z(\omega) \tag{6.21}$$

(using $\mathbf{j}(t) = \mathbf{j}(\omega)e^{i\omega t}$)

Comparing (6.21) with the statement of Ohm's Law ($\mathbf{j}(\omega) = \sigma\mathbf{E}(\omega)$), we obtain the z-component of the conductivity tensor

$$\sigma_{zz} = \frac{ne^2\tau}{m(1 + i\omega\tau)}$$

$$= \frac{\sigma_0}{(1 + i\omega\tau)} \tag{6.22}$$

Similarly, by working with the first two equations of the set (6.18) we can calculate the conductivity in the xy-plane. With the electron instantaneous position and the electric field respectively represented as

$$\mathbf{r}_{xy} = \mathbf{x} + i\mathbf{y}$$

and
$$\mathbf{E}_{xy} = \mathbf{E}_x + i\mathbf{E}_y \tag{6.23}$$

the equation of motion in the xy-plane can be written as

$$m\ddot{\mathbf{r}}_{xy} + \frac{m}{\tau}\dot{\mathbf{r}}_{xy} = -e\mathbf{E}_{xy} + i\omega_c\dot{\mathbf{r}}_{xy} \tag{6.24}$$

The above equation gives

$$m\left[i(\omega - \omega_c) + \frac{1}{\tau}\right]\dot{\mathbf{r}}_{xy} = -e\mathbf{E}_{xy} \tag{6.25}$$

using which we obtain the following expression for the current density

$$\mathbf{j}_{xy}(t) = -ne\dot{\mathbf{r}}_{xy} = \frac{ne^2}{m\left[i(\omega - \omega_c) + \frac{1}{\tau}\right]}\mathbf{E}_{xy}(t)$$

or
$$\mathbf{j}_{xy}(\omega) = \frac{\sigma_0}{\left[1 + i(\omega - \omega_c)\tau\right]}\mathbf{E}_{xy}(\omega) \tag{6.26}$$

This gives the component of the conductivity tensor in xy-plane in the form

$$\sigma^{xy}(\omega) = \frac{\sigma_0}{1 + i(\omega - \omega_c)\tau}$$

$$= \frac{\sigma_0\left[1 - i(\omega - \omega_c)\tau\right]}{1 + (\omega - \omega_c)^2\tau^2} \tag{6.27}$$

The relation (6.27) indicates a likely resonance effect at $\omega = \omega_c$. This is confirmed by the results of measurement of some important properties such as surface absorption, reflectance and other related optical phenomena. The resonance effect is famous as “Cyclotron Resonance’. It may be remarked here that the use of $(e^{-i\omega t})$ in place of $(e^{i\omega t})$ in solution (6.19) is equally valid. It gives resonance for $-\omega = \omega_c$ which refers to the emission process.

The relationship between the current density and the conductivity tensor in three dimensions may be generalized as

$$\begin{bmatrix} \mathbf{j}_x \\ \mathbf{j}_y \\ \mathbf{j}_z \end{bmatrix} = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{bmatrix} \begin{bmatrix} \mathbf{E}_x \\ \mathbf{E}_y \\ \mathbf{E}_z \end{bmatrix} \tag{6.28}$$

Starting with relations (6.18), one readily establishes the following relation involving the real part of the conductivity tensor:

$$\begin{bmatrix} \mathbf{j}_x \\ \mathbf{j}_y \\ \mathbf{j}_z \end{bmatrix} = \frac{\sigma_0}{1+(\omega - \omega_c)^2\tau^2} \begin{bmatrix} 1 & (\omega - \omega_c)\tau & 0 \\ (\omega_c - \omega)\tau & 1 & 0 \\ 0 & 0 & \dfrac{1+(\omega - \omega_c)^2\tau^2}{1+\omega^2\tau^2} \end{bmatrix} \begin{bmatrix} \mathbf{E}_x \\ \mathbf{E}_y \\ \mathbf{E}_z \end{bmatrix} \tag{6.29}$$

Hall Effect

Consider a metal crystal in the form of a strip. As shown in Fig. 6.2, a steady current of density $\mathbf{j}_x$ flows when a static electric field $\mathbf{E}_x$ is impressed along the x-direction in the strip. On switching a static magnetic field **B** over the region of the crystal, the moving electrons are subject to a combination of two forces. One is caused by the magnetic field and it deflects the electrons in the y-direction, collecting them on one of the two parallel faces along this direction. The deposition of electrons on one face develops a small difference of potential (~μV) producing an electric field $\mathbf{E}_y$ along the y-direction. This field creates the other force on electrons, which as per the geometry of the experimental set-up, opposes the force associated with the magnetic field. The net force is expressed as:

$$\mathbf{F} = -e\mathbf{E}_y - e\mathbf{v} \times \mathbf{B} \tag{6.30}$$

called *Lorentz force*.

As soon as the measure of the two oppositely directed force components of Lorentz force become equal, the deflection of electrons stops, culminating into the saturation of the field $\mathbf{E}_y$; popularly known as *Hall field.* The condition of saturation gives:

$$v = \frac{E_y}{B_z} \quad \text{or} \quad \frac{E_y}{j_x B_z} = \left(-\frac{1}{ne}\right) = R_H \tag{6.31}$$

which defines 'Hall coefficient', R_H.

The Hall coefficient is an important parameter needed for the interpretation of the electrical transport in materials. The relation (6.31) follows also from (6.29) if $\mathbf{E}_x$ is made static and $\mathbf{E}_y$ regarded as the Hall field. As such (6.29) assumes the form:

$$\begin{bmatrix} j_x \\ j_y \end{bmatrix} = \frac{\sigma_o}{1+(\omega_c\tau)^2} \begin{bmatrix} 1 & -\omega_c\tau \\ \omega_c\tau & 1 \end{bmatrix} \begin{bmatrix} E_x \\ E_y \end{bmatrix}$$

or

$$j_x = \frac{\sigma_o}{1+(\omega_c\tau)^2}(E_x - \omega_c\tau E_y) \tag{6.32}$$

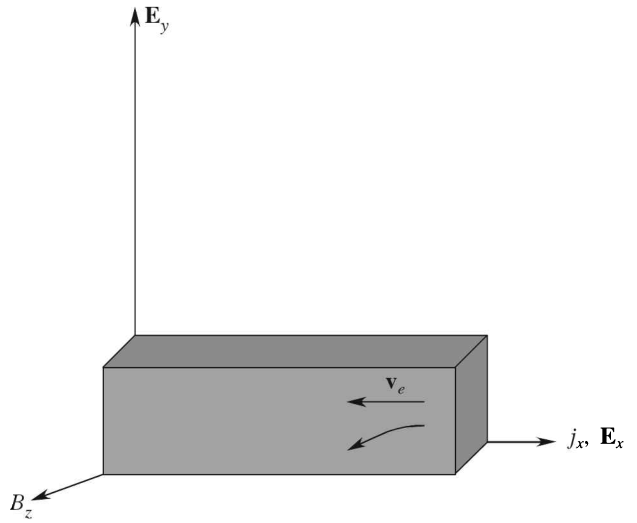

FIG. 6.2 Geometry of the Hall effect. The applied electric field $\mathbf{E}_x$ is along the x-direction. The $\mathbf{E}_y$ denotes the developed Hall voltage. The $\mathbf{v}_e$ represents the electron velocities.

and
$$j_y = \omega_c \tau E_x + E_y \tag{6.33}$$

Since, $j_y = 0$, in the Hall effect geometry,

$$E_x = -\frac{E_y}{\omega_c \tau} \quad \text{[from (6.33)]}$$

Substituting the above in (6.32),

$$j_x = -\frac{\sigma_o}{\omega_c \tau} E_y$$

$$= -\frac{ne}{B_z} E_y$$

Therefore, $\left(-\frac{1}{ne}\right) = \frac{E_y}{j_x B_z} = R_H$ which is same as (6.31), expressing the Hall coefficient.

6.1.2 Thermal Conductivity of Metals

In the Drude model it is assumed that the major contribution to thermal conductivity of metals comes from conduction electrons. This assumption has basis in the general observation that metals are far better conductors of heat than insulators. The participation of metal ions is reflected in the phonon contribution which is neglected in Drude theory on account of its relatively much small measure.

Consider a metal bar whose two ends are maintained at a constant difference of temperature. This situation refers to the steady state when the whole of thermal energy being fed at the hot end is received at the cold end without any net absorption in the bar. Let the temperature gradient along the length of the bar be defined as $-\partial T/\partial \mathbf{x}$, meaning thereby that the temperature decreases down the length being denoted by x. For a small temperature gradient, the thermal current $\mathbf{j}$, which is a measure of the thermal energy flowing per unit sectional area of the bar per unit time, is found to be proportional to the temperature gradient and thus

$$\mathbf{j} = K_{el}\left(-\frac{\partial T}{\partial \mathbf{x}}\right) \tag{6.34}$$

where the constant of proportionality, K_{el}, stands for the electronic thermal conductivity.

Let us confine ourselves to the one-dimensional flow of heat energy in which electrons move parallel to the length of the bar. The calculation of $\mathbf{j}$ proceeds on lines parallel to those discussed in Section 5.3.3 in connection with the phonon current. The main difference between the two cases is that the number of heat carriers in this case (electrons) is conserved. Let us concentrate on a group of n electrons per unit volume. At a point x along the length, half of them ($n/2$) arrive from the hot end and the rest ($n/2$) from the cold end. The electron flowing from the hot end should have met its last collision at $(x - v_x\tau)$ whereas the one from the cold end would have its last collision at $(x + v_x\tau)$. If the thermal energy per electron in thermal equilibrium at T be denoted by $u[T(x)]$, the net thermal current can be written as

$$j = \frac{1}{2}\, n v_x[u(T(x - v_x\tau)) - u(T(x + v_x\tau))] \tag{6.35}$$

where $T(x)$ is the temperature at x and v_x is the x-component of the average electron velocity. In the approximation that the variation in temperature over a mean free path length Λ is small, (6.35) can be expanded about x giving,

$$j = n v_x^2 \tau \,\frac{\partial u}{\partial T}\left(-\frac{\partial T}{\partial x}\right) \tag{6.36}$$

The factor $n(\partial u/\partial T)$ may be replaced by the electron heat capacity per unit volume C_{el} and v_x^2 by $\frac{1}{3}v^2$ in (6.36) to get

$$j = \frac{1}{3}\, C_{el} v^2 \tau\left(-\frac{\partial T}{\partial x}\right) \tag{6.37}$$

Comparing (6.37) with (6.34), we get

$$K_{el} = \frac{1}{3}\, C_{el} v^2 \tau = \frac{1}{3} C_{el} v \Lambda \tag{6.38}$$

where v is the RMS speed of electrons.

The calculation of C_{el} is made by taking the thermal energy per electron at temperature T as $\frac{3}{2}k_B T$, in accordance with the law of equipartition of energy. This gives the heat capacity per electron as $\frac{3}{2}k_B$, and $C_{el} = \frac{3}{2}nk_B$. Furthermore, the value of v^2 is obtained from (6.7). Making these substitutions in (6.38), we get

$$K_{el} = \frac{3}{2}\left(\frac{nk_B^2\tau}{m}\right)T \tag{6.39}$$

Even before the electrons were known, in 1835 Wiedemann and Franz were able to notice that all good conductors of electricity are good conductors of heat too. They established that the ratio of thermal conductivity to electrical conductivity is proportional to the temperature for a large number of metals. Later in 1881 Lorenz (different from Lorentz) observed that the proportionality constant equalling $K/\sigma T$ has almost the same value for most of the common metals. This constant is known as the Lorenz number and the empirical law became famous as the Wiedemann–Franz law. From (6.3) and (6.39), Drude obtained

$$\frac{K}{\sigma} = \frac{3}{2}\left(\frac{k_B}{e}\right)^2 T \tag{6.40}$$

The coefficient of T in (6.40) is a constant. Drude was thus able to derive the Wiedemann–Franz law which is acknowledged as the biggest success of the Drude model. The Lorenz number L as defined in the theory is given by

$$\begin{aligned} L = \frac{K}{\sigma T} &= \frac{3}{2}\left(\frac{k_B}{e}\right)^2 \\ &= 1.11 \times 10^{-8}\ \text{W}\ \Omega\ \text{K}^{-2} \\ &= 1.11 \times 10^{-8}\ (\text{V K}^{-1})^2 \end{aligned} \tag{6.41}$$

But this number is about half the experimental value given by Table 6.2 and Fig. 6.1. Though Drude theory correctly describes the Wiedemann–Franz law, this disagreement points to certain weak links in the theory. However, it is amusing to learn that Drude obtained the correct Lorenz number because of the mistake of getting half the value of σ given by (6.3). The impressive success as it sounded at that time turns out to be misconceived further because of the errors in the calculation of thermal conductivity. The value of the electron heat capacity $\left(\frac{3}{2}nk_B\right)$ used is exactly half the Dulong–Petit value, which is quite large compared to its negligible values observed at room temperature. The values are about a hundred times smaller. Another error appears in the magnitude of electrons thermal velocity which, in fact, is a hundred times larger. It is simply a pleasant coincidence that the two errors almost cancel each other in (6.38), giving an overall success to the Drude model.

The experimental values of thermal conductivity (K) and Lorenz number (L) of some metals at 273 K and 373 K are given in Table 6.2. The observed behaviour of these and of electrical conductivity (σ) for a highly pure sample of copper metal is shown in Fig. 6.1 over a wide range of temperature. It shows that the L-value remains steady above 100 K and keeps falling on cooling below this temperature. As the temperature approaches ~10 K, the value holds on to almost a constant level over the rest of the entire low temperature region. The observations seem to challenge the theoretical basis of Wiedemann-Franz law. Accordingly, we are urged to recall that in the derivation of (6.41), the same τ (the mean free path for that matter) was used for thermal and

Table 6.2 Experimental thermal conductivities* and Lorenz number for some metals

Metal	273 K		373 K	
	Thermal conductivity (W cm^{-1} K^{-1})	*Lorenz number* $(V\ K^{-1})^2$	*Thermal conductivity* (W cm^{-1} K^{-1})	*Lorenz number* $(V\ K^{-1})^2$
Na	1.38	2.12×10^{-8}	melts	
Cu	3.85	2.20	3.82	2.29×10^{-8}
Ag	4.18	2.31	4.17	2.38
Au	3.10	2.32	3.1	2.36
Al	2.38	2.14	2.30	2.19
Fe	0.80	2.61	0.73	2.88
Tl	0.50	2.75	0.45	2.75
Bi	0.09	3.53	0.08	3.35

* From G.W.C. Kaye and T.H. Laby, *Table of Physical and Chemical Constants* (Longmans Green, 1966).

electrical conduction, and hence, got cancelled while taking the ratio of K to σ. But the assumption is legitimate only if the impurity/defect scattering is the dominant resistance mechanism which can happen only at low temperatures and also at high temperatures, as explained by Rosenberg*. Over these temperature ranges, both the electrical and thermal currents are contributed by the flow of electrons across the Fermi surface through a similar process ('horizontal process'). Therefore, Wiedemann–Franz law for metals is valid at low and high temperatures, but fails at intermediate temperatures, as confirmed by the behaviour of Lorenz number in Fig. 6.1. The failure finds explanation in the fact that the inelastic electron-phonon scattering degrades the thermal and electrical currents differently at these temperatures. The law holds good with an accuracy of 5–10% in most of the metals at room temperature.

The τ-independent form of (6.41) would suggest that the Drude theory is capable of accounting for the physical properties that are found τ-independent. Lorentz substantiated this contention for Hall coefficient while working in a refined framework of Drude model.

6.2 LORENTZ MODIFICATION OF THE DRUDE MODEL

Lorentz took up the task of modifying the oversimplified Drude model and constructed his theory on the basis of following points:

1. The assumption that all electrons move with the same thermal velocity is abandoned.
2. The classical Maxwell–Boltzmann velocity distribution is perturbed by the presence of an electric field or a thermal gradient. Both of these tend to displace the equilibrium velocity distribution and distort its symmetry.
3. The approach of Boltzmann transport equation is followed to describe the transport of charge and kinetic energy of electrons by a statistical distribution of mobile electrons constituting the electron gas.

* H.M. Rosenberg; *Low Temperature Solid State Physics* (Clarendon Press, 1963).

To avoid duplication, the derivation of the expression for electrical conductivity will be taken up at a proper stage for the quantum model of the free electron gas, i.e. for the free electron Fermi gas. Nevertheless, we give below the relation as obtained by Lorentz:

$$\sigma_L = \left(\frac{8}{3\pi}\right)^{1/2} \frac{ne^2\Lambda}{(3mk_B T)^{1/2}} \tag{6.42}$$

If we denote the Drude conductivity of (6.8) as σ_D, then

$$\sigma_D = \left(\frac{3\pi}{8}\right)^{1/2} \sigma_L \tag{6.43}$$

$$= 1.09\ \sigma_L$$

From (6.42) and (6.43) we learn that the Lorentz modifications provide neither any noteworthy change in the quantitative measure nor any change in the temperature dependence of electrical conductivity. Lorentz, however, went on to analyze the effect of a uniform magnetic field on current carrying conductors, the Hall effect is known as the Hall effect. The net force on an electron moving with velocity **v** under the action of a static electric field **E** and a uniform magnetic field **B** is

$$\mathbf{F} = -\,e[\mathbf{E} + \mathbf{v} \times \mathbf{B}] \tag{6.44}$$

where **F** is called the Lorentz force.

Lorentz solved the Boltzmann transport equation under the above conditions and obtained the following expression for the Hall coefficient,

$$R_H = -\left(\frac{3\pi}{8}\right)\frac{1}{ne} \tag{6.45a}$$

whereas according to Drude theory,

$$R_H = -\frac{1}{ne} \tag{6.45b}$$

The electron density n can be estimated from (6.45) by feeding the measured value† of Hall coefficient in it. These are given in Table 6.3 together with the atomic concentrations in certain metals. The estimate is consistent with the contribution of one conduction electron per atom in sodium. The result is not so good for silver and is worse for other metals. This points to an otherwise established fact that the free electron approximation is best applicable to alkali metals with the lone loosely bound outermost $3s^1$ electron. The subject is effectively treated by the band theory that also accounts for the observed positive Hall coefficient in certain metals.

Other important consequences of the Lorentz solution of Boltzmann transport equation are:

1. *The Thomson effect.* An electromotive force is developed in conductors between whose two ends a temperature gradient is maintained. This effect is treated in Section 6.8.2.

† Determined from $R_H = \dfrac{E_y}{j_x B_z}$, relation 6.31.

Table 6.3 Hall coefficient and electron density of some metals

Metal	*Hall coefficient* (R_H)* ($m^3\ C^{-1}$)	*Electron density* (n) *deduced from* R_H** (m^{-3})	*Atomic concentration deduced from lattice constant* (m^{-3})
Na	-2.5×10^{-10}	2.9×10^{28}	2.5×10^{28}
Ag	-8.4×10^{-11}	8.8×10^{28}	5.8×10^{28}
Al	-3.0×10^{-11}	2.5×10^{29}	6.0×10^{28}
Cd	$+6.0 \times 10^{-11}$	Negative	4.6×10^{28}
Fe	$+2.5 \times 10^{-11}$	Negative	8.5×10^{28}
Ni	-6.0×10^{-10}	1.2×10^{28}	9.2×10^{28}

* From *Handbook of Physics*, Ed., E.U. Condon and H. Qdishaw (McGraw-Hill, 1958).

** Using $R_H = \dfrac{3\pi}{8ne}$.

2. *The magnetoresistance effect.* The resistivity of a metal increases when placed in a magnetic field. The Drude model does not predict this effect.

Whereas the Thomson effect is exploited in several thermoelectric devices, the magnetoresistance effect is immensely useful in the study of electrical transport properties. The magnetoresistance is defined as the fractional increase in the zero field resistivity when the sample is placed in the region of a magnetic field. In the Lorentz model when the Boltzmann equation is solved without ignoring terms in B of higher powers, the magnetoresistance increases as the square of the magnetic field. Experiments have confirmed this square-law dependence. The study of magnetoresistance has proved of tremendous value in the characterization of semiconductors. The knowledge about its behaviour is very crucial in the study of transformation between the metallic and insulating phases of composites. Some of these have interesting properties making them attractive for applications.

6.3 THE FERMI–DIRAC DISTRIBUTION FUNCTION

As already stated, the Sommerfeld theory differs from the classical theories in the replacement of the Maxwell–Boltzmann statistics by the Fermi–Dirac statistics. The distribution of electron thermal speeds in thermal equilibrium at temperature T is described by the function,

$$f(v) = \frac{1}{\exp\left[\left(\frac{1}{2}\,mv^2 - \mu\right)\Big/k_BT\right] + 1} \tag{6.46}$$

It is more commonly expressed as a function of electron energy,

$$f(\varepsilon) = \frac{1}{\exp\,[(\varepsilon - \mu)/k_BT] + 1} \tag{6.47}$$

The $f(\varepsilon)$ is known as the Fermi–Dirac distribution function. It denotes the probability that an orbital of energy ε be populated in an ideal electron gas in thermal equilibrium. The quantity μ is a function of temperature and is called the ***chemical potential.***

The temperature dependence of $f(\varepsilon)$ may be summarized as follows:

At $T = 0$

$$\begin{aligned} f(\varepsilon) &= 0 \qquad \text{for } \varepsilon > \mu \\ &= 1 \qquad \text{for } \varepsilon < \mu \end{aligned} \tag{6.48a}$$

At $T > 0$

$$f(\varepsilon) = \frac{1}{2} \qquad \text{for } \varepsilon = \mu \tag{6.48b}$$

As $T \to 0$

$$\begin{aligned} \lim_{T \to 0} f(\varepsilon) &= 1 \qquad \text{for } \varepsilon < \mu \\ &= 0 \qquad \text{for } \varepsilon > \mu \end{aligned} \tag{6.48c}$$

The interpretation of (6.48a) asserts that none of the electron levels in a metal at absolute zero can be populated if its energy is greater than the value of the chemical potential of the metal. Thus the energy value of the highest populated level at $T = 0$ coincides with the value of the chemical potential. The level is designated as the ***Fermi level*** and its energy is called the ***Fermi energy***, denoted by ε_F. All the electrons are confined to levels below ε_F at $T = 0$. Its value is decided so as to get the correct number for the total number of particles (here, electrons), using the distribution function (6.47).

However, it is observed that

$$\lim_{T \to 0} \mu = \varepsilon_F \tag{6.49}$$

Also, it is found for metals that to a high degree of precision the chemical potential remains equal to the Fermi energy well up to room temperature. This allows μ to be safely replaced by ε_F in (6.47) while dealing with metals at room temperature and below. This practice, however, should not relent us to ignore the difference of μ from its value at absolute zero in general, since at times it may yield grossly inaccurate results.

The variation of $f(\varepsilon)$ with the change in electron energy ε at certain temperatures is shown in Fig. 6.3. For $(\varepsilon - \varepsilon_F) >> k_B T$, (6.47) reduces to the form

$$f(\varepsilon) \simeq \exp[(\varepsilon_F - \varepsilon)/k_B T] \tag{6.50}$$

which is simply the classical Boltzmann distribution. This indicates that for high values of electron energy the Fermi–Dirac function approaches the classical distribution function (see Fig. 6.3). Therefore, in this limit the choice between the classical and the quantum statistics is meaningless as both give similar results.

The Fermi energy ε_F is often expressed in terms of temperature by the relation,

$$\varepsilon_F = k_B T_F \tag{6.51}$$

where T_F, called the ***Fermi temperature***, turns out to be of the order of tens and thousands of kelvin because the Fermi energy per electron is several electron volts.

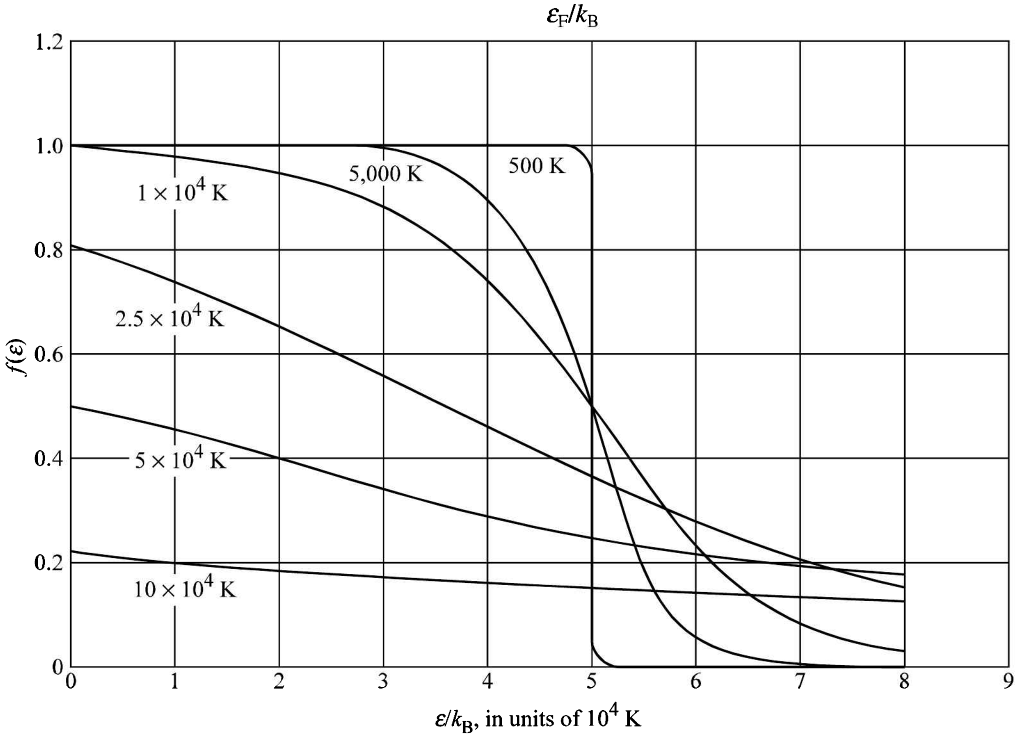

FIG. 6.3 Temperature dependence of the Fermi–Dirac distribution function for a free electron gas in three dimensions, with $T_F = \dfrac{\varepsilon_F}{k_B} = 50{,}000$ K.

6.4 THE SOMMERFELD MODEL

The electron–ion interaction cannot be totally ignored even in metals because the conduction electrons remain confined within the volume of the metal. Since no emission of electrons is observed at room temperature, the potential energy of an electron at rest inside the metal must be lower than that of an electron at rest outside. An electron needs to be excited to a certain level so as to leave the metal and escape to infinity. This level is called the *vacuum level*. The electrons move in the field of the periodic potential of positive ions. Therefore, the potential energy of electrons should be periodic in the crystal lattice. Sommerfeld, however, described the cube of a metal crystal as a three-dimensional potential box with an infinite barrier at the surfaces taking the potential energy as a constant within the box. But it is quite interesting that even such an oversimplified model gives a good agreement with the experiment, though the work function of metals is in the range of 5 eV.

Consider a metal cube of side L. The potential energy of electrons in the metal in the form of an infinite square-well is shown in Fig. 6.4.

Let $V(\mathbf{r})$ denote the potential of an electron at point $\mathbf{r}$ and ε' be its total energy. Then, we can write the time-independent Schrödinger wave equation in the one-electron approximation as

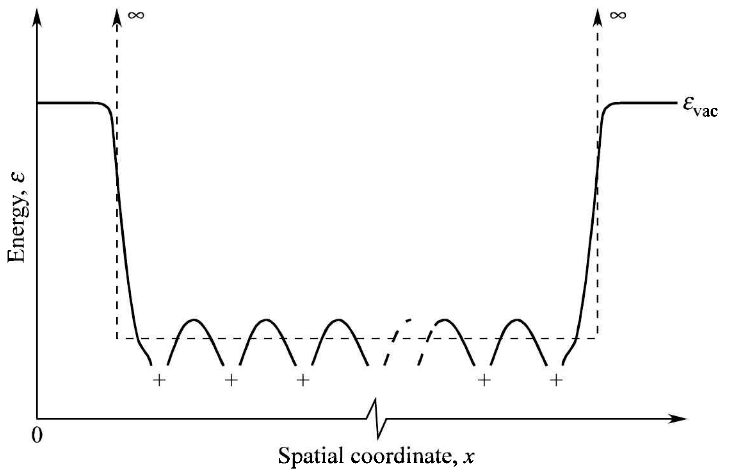

FIG. 6.4 General form of the potential energy of an electron moving in a periodic metallic crystal of positive cores. This is approximated by a square-well potential shown.

$$\left[\frac{\left(-i\hbar\dfrac{d}{d\mathbf{r}}\right)^2}{2m} + V(\mathbf{r})\right]\psi(\mathbf{r}) = \varepsilon'\psi(\mathbf{r}) \tag{6.52}$$

where $-i\hbar(d/d\mathbf{r})$ stands for the linear momentum per electron and $\psi(\mathbf{r})$ is the electron wavefunction.

Sommerfeld defined $V(\mathbf{r})$ as

$$V(x, y, z) = V_0, \quad \text{a constant for } 0 \le x, y, z \le L$$
$$= \infty, \quad \text{otherwise}$$

Let the kinetic energy per electron be denoted by ε, such that $\varepsilon = \varepsilon' - V_0$, which gives

$$-\frac{\hbar^2}{2m}\nabla^2\,\psi(\mathbf{r}) = \varepsilon\,\psi(\mathbf{r}) \tag{6.53}$$

The infinite barrier at the bounding surfaces (x, y, z = 0 and L) keeps the electrons confined within the crystal and gives the following boundary conditions for the wavefunction:

$$\begin{aligned} \psi = 0 \quad \text{for} \quad & x = 0 \text{ and } L; \text{ and } 0 < y, z < L \\ & y = 0 \text{ and } L; \text{ and } 0 < z, x < L \\ & z = 0 \text{ and } L; \text{ and } 0 < x, y < L \end{aligned} \tag{6.54}$$

These fixed boundary conditions require the electrons to remain confined within the potential box. Therefore, the free electron plane waves moving along the x-edge of the cube [exp $(ik_x x)$] have to be reflected back just on reaching the potential barrier. The two oppositely moving plane waves may form standing waves [exp $(ik_x x)$ ± exp $(-ik_x x)$] identified as cosine and sine waves, respectively.

Since only sine waves satisfy the boundary condition (6.54), they serve as the proper solutions to the Schrödinger equation (6.53). The solution is expressed as

$$\psi(r) = A \sin k_x x \sin k_y y \sin k_z z \tag{6.55}$$

where A [$= (2/L)^{3/2}$] is a constant and can be determined normalizing $\psi(r)$ over the potential box. The three Cartesian components of the wavevector **k** are given by

$$\begin{aligned} k_x &= \frac{\pi}{L} n_x \\ k_y &= \frac{\pi}{L} n_y \\ k_z &= \frac{\pi}{L} n_z \end{aligned} \tag{6.56}$$

with $n_x, n_y, n_z = 1, 2, 3, \ldots$

On substituting (6.55) in (6.53), we get the following electron energy states:

$$\varepsilon_{\mathbf{k}} = \frac{\hbar^2}{2m}\left(k_x^2 + k_y^2 + k_z^2\right) \tag{6.57}$$

$$= \frac{\hbar^2 k^2}{2m} \tag{6.57a}$$

$$= n^2 \frac{1}{2m}\left(\frac{h}{2L}\right)^2 \tag{6.57b}$$

with $n_x^2 + n_y^2 + n_z^2 = n^2$.

For an electron moving in the free space, we use (6.57a) to express the energy. The variation of energy with wave vector as plotted in Fig. 6.5 is known as the free electron dispersion curve. But when a free electron moves in a potential box, as in the present case, the electron energy is meaningfully given by (6.57b).

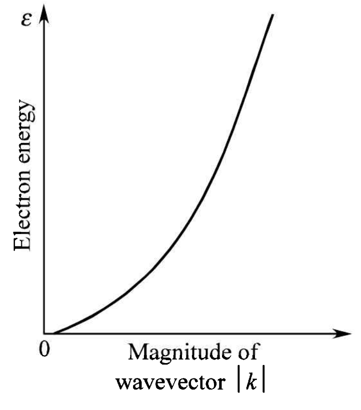

FIG. 6.5 Variation of the free electron energy as a function of the magnitude of its wavevector.

The components (k_x, k_y, k_z) or (n_x, n_y, n_z) together with the spin quantum number m_s are the quantum numbers of the problem. The wavefunctions and the energy states of a free electron in a square-well potential of length L in the x-direction are depicted in Fig. 6.6. The allowed energy values in the three-dimensional space produce constant spherical energy surfaces,

$$\varepsilon_{\mathbf{k}} = \frac{\hbar^2 k^2}{2m} = \text{constant}$$

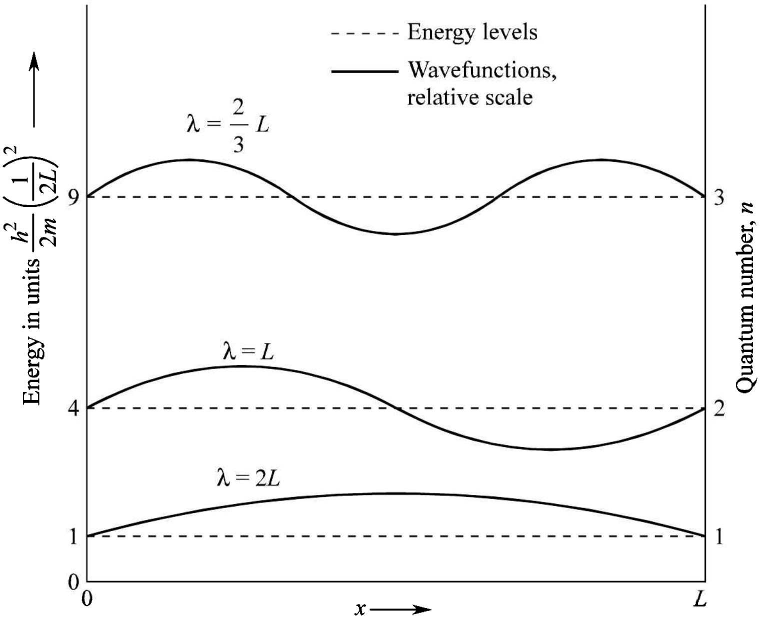

FIG. 6.6 Eigenvalues and eigenfunctions of the first three states of a free electron confined in a potential box of length L.

6.4.1 The Density of States

An insight into the possible values of k_x, k_y, k_z given by (6.56) provides the clue to the determination of density of electron states. We have excluded the zero value of n_x, n_y and n_z since the respective solutions cannot be normalized over the potential box. Further, no new linearly independent solutions are found for the negative values of these integers and as such these values are not considered. Thus for the fixed boundary condition (6.54), the useful possible values of **k** totally lie in the positive octant of the k-space.

In a larger picture, the k-values belonging to the higher electron states may extend to the higher order Brillouin zones. But the periodicity in the k-space ensures the representation of these k-values in the first Brillouin zone, accomplished on translating them by the reciprocal lattice vectors. It implies that all useful values of k are confined within the first Brillouin zone. The depiction of the entire assembly of energy states in the first Brillouin zone is now a norm.

Equations (6.56) show that there is one electron state in volume $(\pi/L)^3$ of the k-space. The principle of calculation of the electron density of states is the same as that for phonons described in Section 5.2.3. Let us calculate the number of states in the thin shell of the octant enclosed between the energy surfaces $\varepsilon(\mathbf{k})$ and $\varepsilon(\mathbf{k}) + d\varepsilon$ [see Fig. (6.7)]. This number, obtained by dividing the volume of this part of the shell by the volume associated with a single state, is

$$\frac{\left(\frac{1}{8}\right)(4\pi k^2 dk)}{\left(\frac{\pi}{L}\right)^3}$$

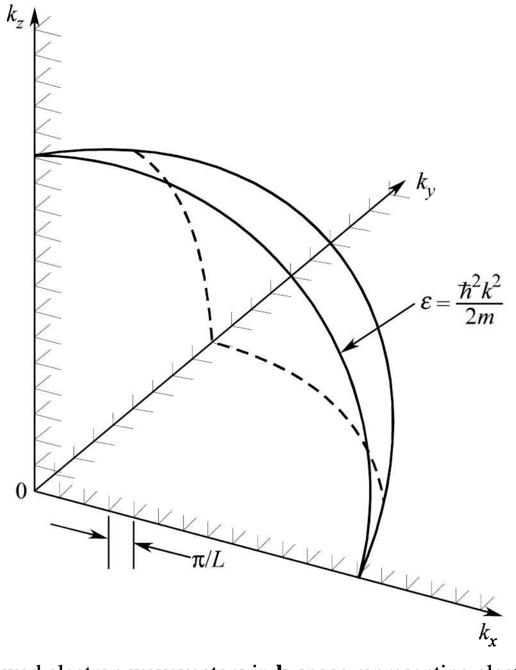

FIG. 6.7 A lattice of the allowed electron wavevectors in **k**-space representing electron states in an infinite square well. Because of the two possible spin values, each point gives two states. The states have a linear separation of π/L and lie completely in one octant for the fixed boundary conditions.

The correct number of possible states is twice this number since the Pauli principle allows two electrons with opposite spins ($\pm$ 1/2) to be accommodated in a single orbital designated by a **k**-value. The two electrons in an orbital are said to be in different states that are degenerate (degenerate states have a common eigenvalue). Therefore, if $D(\varepsilon)$ denotes the density of states,

$$D(\varepsilon)\, d\varepsilon = 2\,\frac{\left(\frac{1}{8}\right)(4\pi k^2 dk)}{\left(\frac{\pi}{L}\right)^3} = \frac{L^3}{\pi^2}\, k^2\, dk$$

$$= \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} \varepsilon^{1/2} d\varepsilon \qquad \left(\text{using } \varepsilon_k = \frac{\hbar^2 k^2}{2m}\right)$$

Therefore,

$$D(\varepsilon) = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} \varepsilon^{1/2} \tag{6.58}$$

where V is the volume of the crystal. We may express the density of states per unit volume of the crystal as

$$\frac{D(\varepsilon)}{V} = \frac{1}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} \varepsilon^{1/2} \tag{6.59}$$

The density of states as a function of energy is graphically represented in Fig. 6.8.

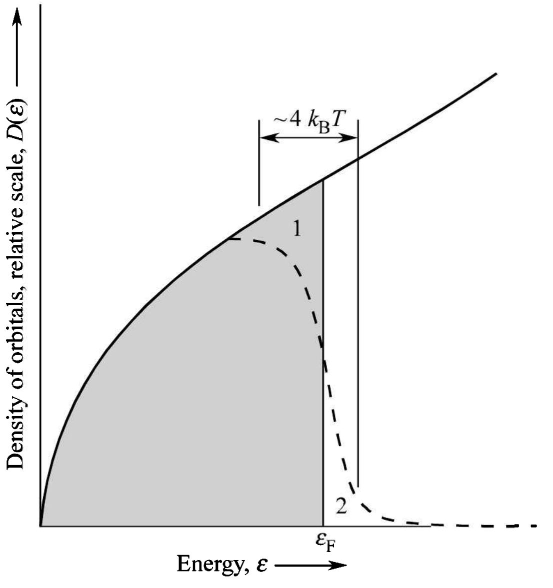

FIG. 6.8 Variation of the density of single-particle states as a function of temperature for a free electron gas in three dimensions. For $k_B T < \varepsilon_F$, the density of the filled orbitals $D(\varepsilon)\, f(\varepsilon, T)$ at finite temperature is given by the dashed curve. The shaded area represents the filled orbitals at 0 K. On heating to temperature T, the average energy increases and the electrons from region 1 (below ε_F) are thermally excited to region 2 (above ε_F).

It is left as an exercise for the reader to show that the periodic boundary conditions,

$$\psi(x + L, y + L, z + L) = \psi(x, y, z)$$

that yield plane wave solutions to the Schrödinger equation (6.53), lead to the same expression for the density of states as given by (6.58).

6.4.2 The Free Electron Gas at 0 K

The state of the free electron gas at the absolute zero temperature forms the basis for studying the properties of metals. We know that at this temperature all the electrons are filled in orbitals below the Fermi level, ε_F. Using this concept, we calculate the total number of electrons,

$$N = \int_0^{\varepsilon_F} D(\varepsilon) f(\varepsilon)\, d\varepsilon$$

$$= \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\int_0^{\varepsilon_F} \varepsilon^{1/2}\, d\varepsilon \qquad (\text{for } \varepsilon < \varepsilon_F,\ f(\varepsilon) = 1)$$

or

$$N = \frac{V}{3\pi^2}\left(\frac{2m\varepsilon_F}{\hbar^2}\right)^{3/2} \tag{6.60}$$

The density of states at the Fermi level can be quickly obtained in terms of the total number of electrons if we rewrite (6.60) as

$$\ln N = \frac{3}{2}\ln \varepsilon_F + \text{a constant} \tag{6.61}$$

Differentiating it, we get

$$\frac{dN}{d\varepsilon_F} = \frac{3N}{2\varepsilon_F} = D(\varepsilon_F) \tag{6.62}$$

If we denote the number of electrons per unit volume by n,

$$n = \frac{N}{V} = \frac{1}{3\pi^2}\left(\frac{2m\varepsilon_F}{\hbar^2}\right)^{3/2} \tag{6.63}$$

giving the Fermi energy ε_F as

$$\varepsilon_F = \frac{\hbar^2}{2m}(3\pi^2 n)^{2/3} \tag{6.64}$$

Thus the Fermi energy may be estimated using (6.64) with the knowledge of the electrons density, n. Since metals have about 10^{28} electrons per m^3, the magnitude of ε_F is several eV per electron. From (6.51) we get a Fermi temperature, T_F in the range of tens of thousands of kelvin which is enormously higher than the room temperature. On this scale of temperature the room temperature is far more closer to 0 K. Furthermore, experiments show that most of the properties of metals at room temperature are not much different than those at 0 K. This permits us to use the value of the density of states near the Fermi level for all calculations by replacing ε with ε_F in (6.58).

The size of ε_F is also measured in terms of the Fermi wavevector k_F, whose magnitude equals the radius of the spherical Fermi energy surface in the three-dimensional k-space. From (6.64), we have

$$k_F = (3\pi^2 n)^{1/3} \tag{6.65}$$

The electrons move on the Fermi surface with a constant speed, v_F expressed as

$$v_F = \frac{\hbar k_F}{m} = \frac{\hbar}{m}(3\pi^2 n)^{1/3} \tag{6.66}$$

6.4.3 Energy of Electron Gas at 0 K

This estimate is invaluable to several calculations involving free electrons. The energy per unit volume is expressed as

$$U_0 = \frac{1}{V}\int_0^{\varepsilon_F} \varepsilon\, D(\varepsilon)\, d\varepsilon \tag{6.67}$$

Substituting $D(\varepsilon)$ from (6.58), we obtain

$$U_0 = \frac{\varepsilon_F^{5/2}}{5\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} \tag{6.68}$$

It is more useful to know the energy per electron, also referred to as the average kinetic energy per free electron. We obtain this value by dividing (6.68) by (6.63). Therefore,

$$\text{Average kinetic energy per electron} = \frac{U_0}{n} = \frac{3}{5}\varepsilon_F \tag{6.69}$$

EXAMPLE 6.1 Derive expressions for the electron density of states in a two-dimensional and in a one-dimensional free electron Fermi gas at absolute zero temperature. Compare the forms of variation with that in the three-dimensional case.

Solution: The number of electrons contained within the constant energy contours ε and $\varepsilon + d\varepsilon$ in a two-dimensional crystal in reciprocal space can be expressed as:

$$D(\varepsilon)\, d\varepsilon = 2\frac{S}{(2\pi)^2}\int_{l_\varepsilon} d^2k = 2\frac{S}{(2\pi)^2}\int_{l_\varepsilon} dl_\varepsilon \frac{dk}{d\varepsilon}\, d\varepsilon \quad \text{[in analogy with (5.24) for phonons]}$$

where, S is the area and l_ε is the contour of constant energy.

$$d\varepsilon = \frac{\partial\varepsilon}{\partial k_x}\cdot dk_x + \frac{\partial\varepsilon}{\partial k_y}\cdot dk_y = |\nabla_k \varepsilon|\, dk$$

or

$$\frac{dk}{d\varepsilon} = \frac{1}{\nabla_k \varepsilon}$$

Therefore,

$$D(\varepsilon)^{2D} = 2\frac{S}{(2\pi)^2}\int_{l_\varepsilon} \frac{dl_\varepsilon}{\nabla_k \varepsilon}$$

In a two-dimensional electron gas $\nabla_k \varepsilon \left(= \dfrac{\hbar^2 k}{m} \right)$ is a constant along a circular path of radius k. This gives:

$$D(\varepsilon)^{2D} = 2\frac{S}{(2\pi)^2} \cdot 2\pi k \cdot \frac{m}{\hbar^2 k} = \frac{mS}{\pi\hbar^2} = \text{a constant (say, } C_2\text{)}$$

At absolute zero for an N-electron system, the constant can be obtained in terms of Fermi energy by using the normalization:

$$\int_0^{\varepsilon_F} D(\varepsilon)^{2D} d\varepsilon = N$$

or

$$C_2 \int_0^{\varepsilon_F} d\varepsilon = N$$

or

$$C_2 = D(\varepsilon)^{2D} = \frac{N}{\varepsilon_F}$$

where ε_F is the Fermi energy at absolute zero.

In one dimension, the integration contour extends to points $\pm k$. Therefore, the free electron density of states reduces to:

$$D(\varepsilon)^{1D} = 2\frac{L}{2\pi} \cdot 2 \cdot \frac{m}{\hbar^2 k} = \frac{L(2m)^{1/2}}{\pi\hbar} \varepsilon^{-1/2} = C_1 \varepsilon^{-1/2}$$

Normalizing at absolute zero,

$$\int_0^{\varepsilon_F} D^{1D}(\varepsilon)\, d\varepsilon = N \quad \text{or} \quad C_1 \int_0^{\varepsilon_F} \varepsilon^{-1/2}\, d\varepsilon = N \quad \text{or} \quad C_1 = \frac{N}{2\varepsilon_F^{1/2}}$$

Therefore,

$$D(\varepsilon)^{1D} = \left(\frac{N}{2\varepsilon_F^{1/2}} \right) \varepsilon^{-1/2}$$

Combining (6.58) and (6.60), we get the expression for three-dimensional gas as:

$$D^{3D}(\varepsilon) = \left(\frac{3N}{2\varepsilon_F^{3/2}} \right) \varepsilon^{1/2}$$

The change in the form of variation of the electron density of states occurring with the change in dimensionality is reflected in the tracings shown in Fig. Ex. 6.1.

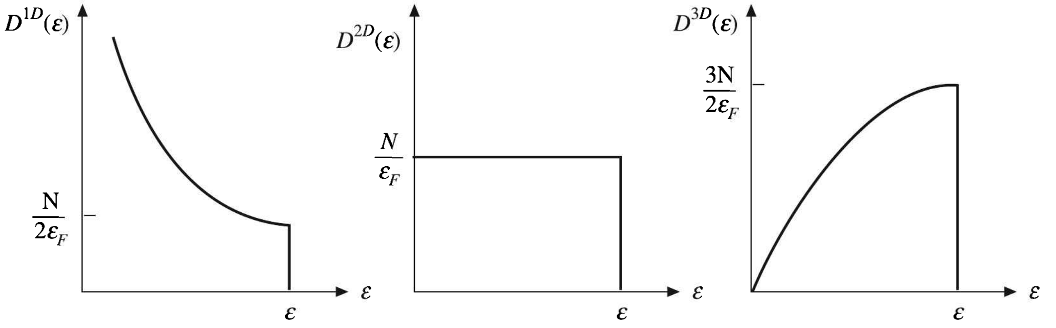

FIG. Ex. 6.1

EXAMPLE 6.2 Consider a two-dimensional hexagonal lattice of lattice constant a = 2.46 Å at 0 K, with one electron per unit cell. Treating electrons as free within the two-dimensional plane, calculate the Fermi energy of the electron gas.

Solution: The primitive vectors of the direct lattice are at 60° angle, while the angle between the respective vectors of the reciprocal lattice is 120° (see Ex. 3.1). Since all the allowed k-values can be confined within the first Brillouin zone, all electrons can be considered as enclosed within the first Brillouin zone. If there are N primitive cells, the number of allowed values is also N. Therefore, the density of states (per unit area) in the reciprocal space is:

$$\frac{N}{\boldsymbol{a}_1^* \times \boldsymbol{a}_2^*} = \frac{N}{\left(\dfrac{4\pi}{a\sqrt{3}}\right)^2 \sin 120°} = \frac{Na^2}{8\pi^2}\sqrt{3}$$

Each state can accommodate two electrons with opposite spins and all electrons are enclosed within the Fermi circle at absolute zero.
Therefore,

$$2 \times \pi k_F^2 \times \frac{Na^2}{8\pi^2}\sqrt{3} = N \quad \text{or} \quad k_F = \frac{1}{a}\sqrt{\frac{4\pi}{\sqrt{3}}}$$

and the Fermi energy,

$$\varepsilon_F = \frac{\hbar^2 k_F^2}{2m} = 4.58 \text{ eV}$$

6.5 THE ELECTRON HEAT CAPACITY

The most glaring thermal anomaly concerning the electron heat capacity, as encountered in the Drude model, was resolved by Sommerfeld with the use of quantum statistics. He applied Fermi–Dirac statistics to study the properties of the free electron gas. The density of states that describes the number of energy states per unit energy interval is shown in Fig. 6.8 as a function of energy. Sommerfeld pointed out that the smeared out region of the distribution function has a width of the order of $4k_BT$. According to Fig. 6.8 the electrons in the energy range $2k_BT$ below the Fermi level undergo excitation to cross it when the metal is heated from absolute zero to a temperature T. The electrons in still deeper levels are unable to scatter to a slightly different energy on account of all levels of comparable energy being already occupied with no room for additional occupancy as dictated by the Pauli principle. From Fig. 6.8 we find that only the uppermost electrons equalling a fraction of roughly $(4k_BT/\varepsilon_F)$ are affected by the presence of a field gradient (thermal or electrical).

Sommerfeld thus concluded that electrons occupying states only close to the Fermi level ε_F contribute to most of the properties of metals. We can make a rough estimate of the electron heat capacity based on the above picture. Taking the rise in thermal energy per electron as k_BT on heating the system from 0 K to T K, the change in the net thermal or internal energy per unit volume may be given by

$$U = n\left(\frac{4k_BT}{\varepsilon_F}\right)\cdot k_BT$$

$$= \frac{4nk_BT^2}{T_F} \qquad (\text{since } \varepsilon_F = k_BT_F) \tag{6.70}$$

Therefore, the electron heat capacity C_{el} is written as

$$C_{el} = \frac{\partial U}{\partial T} = 8nk_B\left(\frac{T}{T_F}\right) \tag{6.71}$$

It is definitely a very small number since T_F is around 10^5 K. The order of magnitude is consistent with the experimental observation that there is hardly a deviation from the Duloug–Petit value of heat capacity at room temperature. This confirms absurdity of the Drude's estimate of $\frac{3}{2}nk_B$ which is exactly half the Dulong–Petit value.

Now, we take up a proper calculation of the heat capacity contributed by the conduction electrons. The net increase in internal energy per unit volume as the temperature rises from 0 K to a temperature T is written as

$$U = \int_0^\infty \varepsilon D(\varepsilon) f(\varepsilon)\, d\varepsilon - \int_0^{\varepsilon_F} \varepsilon D(\varepsilon)\, d\varepsilon \tag{6.72}$$

with the total number of electrons per unit volume given by

$$n = \int_0^\infty D(\varepsilon) f(\varepsilon)\, d\varepsilon \tag{6.73a}$$

So, we write

$$\varepsilon_F \,.\, n = \varepsilon_F \int_0^\infty D(\varepsilon) f(\varepsilon)\, d\varepsilon \tag{6.73b}$$

On differentiating (6.72) and (6.73b), we get

$$C_{el} = \frac{\partial U}{\partial T} = \int_0^\infty \varepsilon D(\varepsilon)\, \frac{\partial f}{\partial T}\, d\varepsilon$$

$$0 = \varepsilon_F \frac{\partial n}{\partial T} = \int_0^\infty \varepsilon_F D(\varepsilon)\, \frac{\partial f}{\partial T}\, d\varepsilon$$

Subtracting the latter from the former of the above equations, we get

$$C_{el} = \int_0^\infty (\varepsilon - \varepsilon_F)\, D(\varepsilon)\, \frac{\partial f}{\partial T}\, d\varepsilon \tag{6.74}$$

An examination of Fig. 6.3 and Fig. 6.8 shows that at temperatures of interest the derivative $\partial f/\partial T$ has significant values in the region of $\pm\, 2k_BT$ about ε_F. The variation of $D(\varepsilon)$ in this region is not much and, therefore, we can use $D(\varepsilon_F)$ for $D(\varepsilon)$ in the first approximation getting,

$$C_{el} = D(\varepsilon_F) \int_0^\infty (\varepsilon - \varepsilon_F)\, \frac{\partial f}{\partial T}\, d\varepsilon \tag{6.75}$$

Differentiating the Fermi–Dirac function (6.47), we obtain

$$\frac{\partial f}{\partial T} = \frac{\varepsilon - \varepsilon_F}{k_B T^2}\, \frac{\exp\,[(\varepsilon - \varepsilon_F)/k_BT]}{\{\exp\,[(\varepsilon - \varepsilon_F)/k_BT + 1]\}^2} \tag{6.76}$$

Putting $\dfrac{\varepsilon - \varepsilon_F}{k_BT} = x$, we get

$$C_{el} \approx k_B^2\, TD\,(\varepsilon_F) \int_{-\varepsilon_F/k_BT}^{\infty} \frac{x^2\, e^x}{(e^x + 1)^2}\, dx \tag{6.77}$$

The e^x factor in the integrand is negligible for $x \le -\,\varepsilon_F/k_BT$, i.e. at low temperatures. Therefore, the lower limit of integration may be safely extended to $-\,\infty$ transforming the integral into a standard form:

$$\int_{-\infty}^{\infty} \frac{x^2 e^x}{(e^x + 1)^2}\, dx = \frac{\pi^2}{3} \tag{6.78}$$

When the integral in (6.77) is substituted with this value, obtained from tables of standard integrals, we get

$$C_{\text{el}} = \frac{\pi^2}{3} D(\varepsilon_F)\, k_B^2 T \tag{6.79}$$

Using (6.62), we express the density of states per unit volume at Fermi energy as

$$D(\varepsilon_F) = \frac{3n}{2\varepsilon_F} \tag{6.80}$$

Substituting $D(\varepsilon_F)$ from (6.80) in (6.79), we obtain

$$C_{\text{el}} = \frac{\pi^2}{2}\,\frac{nk_B^2 T}{\varepsilon_F} = \frac{\pi^2}{2}\, nk_B \left(\frac{T}{T_F}\right) \tag{6.81}$$

The measured values of electron heat capacity of metals at low temperature are in consistency with the magnitude and the linear temperature dependence expressed by (6.81). Also, it can be checked that with the use of (6.81) for the value of heat capacity and the Fermi velocity v_F for the electrons thermal speed, we get a Lorenz number $K/\sigma T$ which is in excellent agreement with the experimental value. It is another improvement on the Drude theory. It may, however, be further observed that with the use of Fermi–Dirac statistics the heat capacity has been degenerated or degraded from its large classical value by a factor of about $(\varepsilon_F/3k_BT)$. It is in this sense that the free electron gas is referred to as the ***degenerate electron gas***. We will see that the use of ***degenerate*** and ***non-degenerate*** in respect of the semiconductors has a different meaning.

At low temperatures below the Debye temperature and the Fermi temperature, the heat capacity of a metal is expressed as

$$C_V = \gamma T + \alpha T^3 \tag{6.82}$$

where the first term represents the electron contribution and the second term is the usual Debye term contributed by phonons. The phonon or ionic contribution dominates at high temperature but drops spectacularly at low temperatures where it becomes even smaller than the electron contribution. In metals at low temperatures the linear term dominates. The parameters γ and α can be determined from the conventional graph between C_V/T and T^2 using the measured values of C_V. The graph is a straight line and drawn for copper in Fig. 6.9. The intercept with the ordinate axis is the measure of γ whereas the slope gives the value of α. The experimental values of γ together with their ratio to the theoretical values for a number of metals are given in Table 6.4.

We observe a reasonably good agreement for most of the metals shown in Table 6.4 except for the transition metals, Mn, Co and Ni where the estimates from the free electron model show large deviations. This can be interpreted as a result of the strong localized nature of d-electrons occupying a partially filled shell. Since there is little overlap among electron waves, the problem cannot be solved in the free electron model. The solution is provided by the band theory which is the subject of the next two chapters. The point that can be appreciated here is the irrelevance of using the electron rest mass for calculations because the d-electrons behave as heavier ones on account of their tight binding. This fact is effectively considered in the band theory.

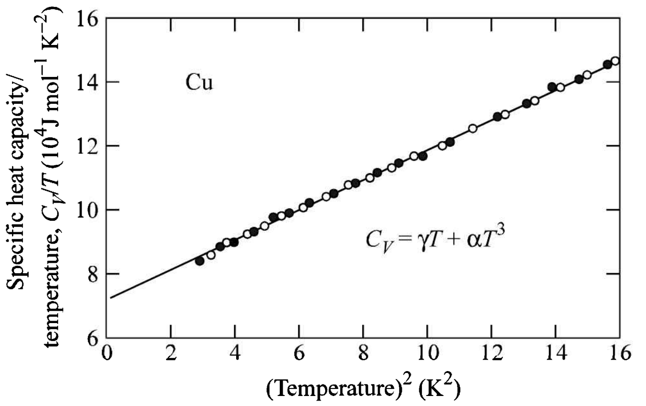

FIG. 6.9 Variation of C_V/T as a function of T^2 for copper. Two sets of experimental points refer to two separate measurements. [After C.A. Bailey, P.L. Smith, *Phys. Rev.*, 114, 1010 (1959).]

Table 6.4 Comparison of the experimental and free electron values of heat capacity constant γ for some metals

Metal	γ_{exp} (mJ mol^{-1} K^{-2})	$\gamma_{exp}/\gamma_{theo.}$
Li	1.7	2.3
Na	1.7	1.5
K	2.0	1.1
Cu	0.69	1.37
Ag	0.66	1.02
Au	0.73	1.14
Al	1.35	1.6
Fe	4.98	10.0
Co	4.98	10.3
Ni	7.02	15.3
Zn	0.64	0.85
Pb	2.98	1.97

6.6 THE SOMMERFELD THEORY OF ELECTRIC CONDUCTION IN METALS

Sommerfeld followed Lorentz approach for calculating the electrical conductivity of metals. This method is based on the use of Fermi–Dirac distribution function for solving the Boltzmann transport equation. A complete rigorous treatment of electrical conduction following this approach is fairly lengthy and complicated. A simplified treatment is outlined in this section to explain the principles involved.

In the absence of a field or thermal gradient, the energy and momentum distribution of the free electron gas is described by the equilibrium distribution function,

$$f^0(p) = \frac{1}{\exp\left[\left(\frac{p^2}{2m} - \varepsilon_F\right)\Big/k_B T\right] + 1} \tag{6.83}$$

As a simplification it is assumed that the metal is homogeneous, so that f^0 is independent of the spatial coordinates. The application of any field deforms this function leading to a changed function. In transport phenomena there are two opposite mechanisms that compete with each other. In electric conduction, the electric field distorts the distribution function from its equilibrium form whereas at the same time the scattering of electrons by phonons and defects tries to restore the equilibrium form of the function. If there is no thermal gradient in the system, the net time rate of change of the distribution function can be expressed as

$$\frac{df}{dt} = \left(\frac{\partial f}{\partial t}\right)_{\text{field}} + \left(\frac{\partial f}{\partial t}\right)_{\text{scatt}} \tag{6.84}$$

where the field term may be treated in general sense so as to include the effect of both electric and magnetic fields depending on the situation. In the presence of a thermal gradient there is an additional term on the RHS of (6.84) contributed by the diffusion process and (6.84) is rewritten as

$$\frac{df}{dt} = \left(\frac{\partial f}{\partial t}\right)_{\text{field}} + \left(\frac{\partial f}{\partial t}\right)_{\text{scatt}} + \left(\frac{\partial f}{\partial t}\right)_{\text{diff.}} \tag{6.85}$$

The steady state condition requires the net rate of change of the function to vanish, i.e. $df/dt = 0$, giving

$$\left(\frac{\partial f}{\partial t}\right)_{\text{field}} + \left(\frac{\partial f}{\partial t}\right)_{\text{scatt}} + \left(\frac{\partial f}{\partial t}\right)_{\text{diff.}} = 0 \tag{6.86}$$

The above equation is the required Boltzmann transport equation. The steady state must not be confused with the equilibrium state that refers to the state when no field or thermal gradients are present.

Now, let us apply (6.86) to electronic conduction in a metal assuming that all points within its volume remain at a constant temperature. Then, (6.84) reduces to the form

$$\left(\frac{\partial f}{\partial t}\right)_{\text{field}} + \left(\frac{\partial f}{\partial t}\right)_{\text{scatt}} = 0 \tag{6.87}$$

Under the influence of the electric field $\mathbf{E}_x$ that acts along the negative x-direction, the velocity distribution of electrons would be drifted towards the positive x-direction. The magnitude of drift velocity v_x depends on the strength of the field. The drift of the spherical Fermi distribution of velocities is shown in Fig. 6.10. As a result the equilibrium distribution f^0 approaches a deformed distribution f. Again, on the ground of simplicity we consider the consequences of a uniform field $\mathbf{E}_x$, so that the spatial derivative of $(f - f^0)$ is zero.

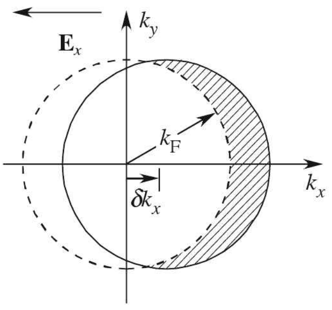

FIG. 6.10 The effect of a steady electric field $\mathbf{E}_x$ on the k-space distribution of quasi-free electrons. The Fermi distribution of equilibrium distribution (dashed) centred at (0, 0, 0) is displaced in the stationary state by an amount $\delta k_x = -e\tau E_x/\hbar$.

The force on an electron owing to the electric field **E** along the x-direction is

$$-e\mathbf{E}_x = \frac{\partial \mathbf{p}_x}{\partial t} \tag{6.88}$$

where $\mathbf{p}_x$ denotes the electron's linear momentum along the x-direction.

In view of (6.88), we have

$$\left(\frac{\partial f}{\partial t}\right)_{\text{field}} = \frac{\partial \mathbf{p}_x}{\partial t} \cdot \frac{\partial f}{\partial \mathbf{p}_x} = (-e\mathbf{E}_x)\frac{\partial f}{\partial \mathbf{p}_x} \tag{6.89}$$

The structure of the scattering term in (6.87) is quite complex and cannot be discussed here in completeness as the related theory involves concepts at an advanced level. This term is modelled in the so-called relaxation time approximation. Lorentz made the postulate that the rate at which f returns to the equilibrium distribution f^0 because of scattering is proportional to the deviation of f from f^0, i.e.

$$\left(\frac{\partial f}{\partial t}\right)_{\text{scatt}} \propto f(\mathbf{p}) - f^0(\mathbf{p})$$

which is reduced to

$$\left(\frac{\partial f}{\partial t}\right)_{\text{scatt}} = \frac{f(\mathbf{p}) - f^0(\mathbf{p})}{\tau} \tag{6.90}$$

It should be noted that the relaxation time τ is defined for each location in the momentum space. Assuming that f does not depend on position (i.e. $\nabla_{\mathbf{r}} f = 0$) and using (6.87), (6.89) and (6.90), we can express the stationary non-equilibrium distribution function by

$$f(\mathbf{p}) = f^0(\mathbf{p}) + eE_x\tau\left(\frac{\partial f}{\partial \mathbf{p}_x}\right) \tag{6.91}$$

The problem is now reduced to solving this differential equation for f. For this, we make use of the Heisenberg uncertainty principle relating uncertainties in the position (x, y, z) and the momentum (p_x, p_y, p_z) giving

$$\Delta x \cdot \Delta y \cdot \Delta z \simeq \frac{h^3}{\Delta p_x \ \Delta p_y \ \Delta p_z} \tag{6.92}$$

where h is the Planck constant.

Since there is one state in volume $\Delta x \cdot \Delta y \cdot \Delta z$, the number of states per unit volume of the crystal in the momentum range $p + dp$ and $p = \dfrac{\Delta p_x \cdot \Delta p_y \cdot \Delta p_z}{h^3} \times 2$.

A factor of 2 is brought in, because of the two spins of the electron (± 1/2). The values of spin create two degenerate states from a single orbital. The electron density that expresses the total number of states per unit volume in the full allowed range of momenta may thus be written as

$$n = \frac{2}{h^3} \iiint f \, dp_x \, dp_y \, dp_z \tag{6.93}$$

The electric current density along the x-direction is usually written as

$$j_x = -\, nev_x$$

$$= -\frac{2e}{h^3} \iiint \left[f^0 + eE_x \tau \left(\frac{\partial f}{\partial p_x} \right) \right] v_x dp_x dp_y dp_z \qquad \text{[using (6.91) and (6.93)]}$$

$$= -\frac{2e^2}{h^3} E_x \iiint \tau \left(\frac{\partial f}{\partial p_x} \right) v_x dp_x dp_y dp_z \tag{6.94}$$

since f^0 is spherically symmetric and does not contribute to the current.

The volume element $dp_x \, dp_y \, dp_z$ in (6.94) equals the volume of a thin spherical shell in the momentum space centred at the sphere of radius p, i.e.

$$dp_x dp_y dp_z = 4\pi p^2 dp \tag{6.95}$$

Further,

$$\frac{\partial f}{\partial p_x} = \frac{\partial f}{\partial \varepsilon} \cdot \frac{\partial \varepsilon}{\partial p_x} \tag{6.96}$$

with the electron energy

$$\varepsilon = \frac{p^2}{2m} = \frac{p_x^2 + p_y^2 + p_z^2}{2m}$$

which gives

$$\frac{\partial f}{\partial p_x} = \left(\frac{\partial f}{\partial \varepsilon} \right) v_x \tag{6.97}$$

Making use of the relations (6.95) to (6.97), j_x from (6.94) can be expressed as

$$j_x = \frac{8\pi e^2 E_x}{h^3} \int_0^\infty \tau v_x^2 \left(-\frac{\partial f}{\partial \varepsilon}\right) m(2m\varepsilon)^{1/2}\, d\varepsilon \tag{6.98}$$

In order to carry out the integration we make the following assumptions to simplify the treatment:

(i) τ depends only on the magnitude of the electron velocity and not on the direction of motion.

(ii) For small fields,

$$\frac{\partial f}{\partial \varepsilon} = \frac{\partial f^0}{\partial \varepsilon}$$

and

$$v_x^2 \simeq \frac{1}{3} v^2$$

(iii) Since $-\partial f/\partial \varepsilon$ has an appreciable value only in an energy range of a few $k_B T$ about the Fermi level ε_F (see Fig. 6.11), the values of τ and ε that matter in calculations are those at the Fermi energy (ε_F). Therefore, to a good approximation τ and ε under the integral sign may be replaced by $\tau(\varepsilon_F)$ and ε_F in that order making them thus independent of energy.

Applying these assumptions, (6.98) is transformed to:

$$j_x = \frac{16\pi e^2 E_x \tau(\varepsilon_F)\, \varepsilon_F^{3/2}}{3h^3} (2m)^{1/2} \int_0^\infty \left(-\frac{\partial f^0}{\partial \varepsilon}\right) d\varepsilon \tag{6.99}$$

Using (6.64) and $\int_0^\infty \left(-\frac{\partial f^0}{\partial \varepsilon}\right) d\varepsilon = 1$, we have

$$j_x = \frac{ne^2 \tau(\varepsilon_F)}{m} E_x$$

or

$$j_x = \sigma E_x \tag{6.100}$$

where

$$\sigma = \frac{ne^2 \tau(\varepsilon_F)}{m} \tag{6.101}$$

The quantity σ is called the electrical conductivity. The relation (6.101) is similar in form to the Drude's relation (6.3). The two differ in the way the mean free time τ is defined. In Sommerfeld

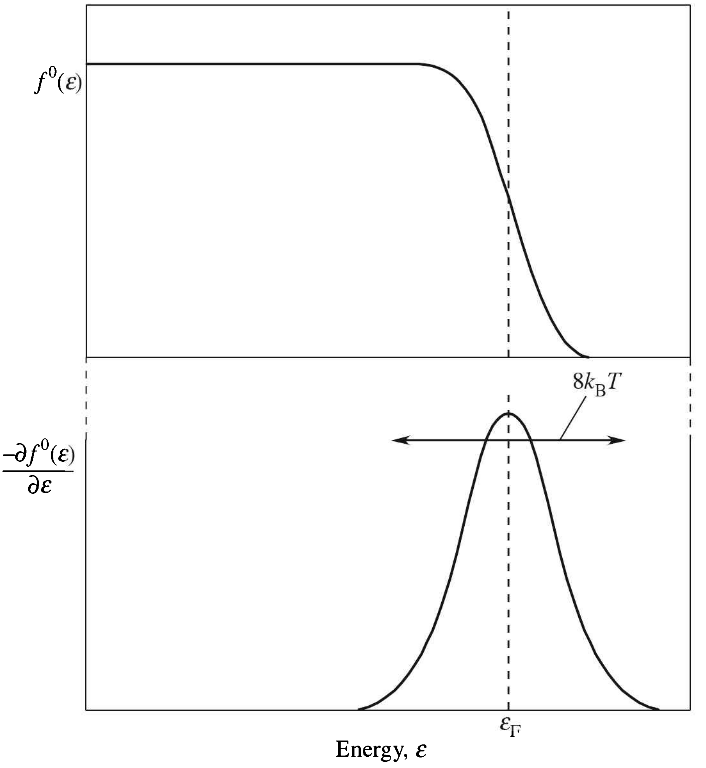

FIG. 6.11 Variation in the Fermi–Dirac distribution function (the occupancy factor) and in its derivative with change in energy.

model it is defined as the ratio $\left(\frac{\Lambda_F}{v_F}\right)$, where Λ_F is the mean free path and v_F the electron speed at the Fermi level. With this definition, the conductivity is written as

$$\sigma = \frac{ne^2\Lambda_F}{mv_F} \tag{6.102}$$

Unfortunately, the observed magnitude and the temperature dependence of conductivity (see Table 6.1 and Fig. 6.1) are not consistent with (6.102). Though Sommerfeld used the quantum statistics, he continued to work in the Lorentz model assuming that electrons made elastic collisions with ion cores, which is far from being true. Therefore, the theory needed extension beyond this assumption to explain the observed behaviour of conductivity. The variation of σ in (6.102) is determined by the change in Λ_F since the magnitude of v_F is almost temperature independent. When the assumption of electron–ion elastic collision is abandoned and the effect is replaced by the electron–phonon scattering, the correct temperature dependence of the mean free path is obtained. At temperatures well above the Debye temperature, the phonon population for all frequencies is proportional to the temperature. This gives T^{-1} dependence to the mean free path and the conductivity in agreement with the experiment. The dependence at low temperatures is, however, T^{-5}

(Bloch–Grüneisen law). The two temperature dependences refer to the different scattering mechanisms active in the two extreme temperature ranges. The plateau observed at low temperatures in the electrical conductivity and the Lorenz number curves of Fig. 6.1 is contributed almost totally by the scattering of electrons from impurities and static crystal imperfections. From discussions in Section 6.7 it will be clear that this scattering is temperature independent. In the high temperature range the electron–phonon collisions dominate over other processes and account alone for the observed behaviour.

Nevertheless, the Sommerfeld model has a unique status in the theory of metals since it identified for the first time the difference between a large number of 'free electrons' and the much smaller number of 'conduction electrons'.

6.6.1 The Hall Coefficient (R_H)

The calculation of the Hall coefficient by Sommerfeld theory is more of academic interest as the estimates from Lorentz theory are also consistent with the measured values. The use of quantum statistics is of no advantage simply because the expression for Hall coefficient does not finally involve the mean free time.

A proper solution of the Boltzmann transport equation in the above problem is tedious and complex and as such is not given here. We straightaway quote the solution. Under the combined influences of $\mathbf{E}_x$ and $\mathbf{E}_y$, the general steady-state solution to the first order in $\mathbf{B}$ is

$$\mathbf{E} = \frac{\mathbf{j}}{\sigma} + \frac{\omega_c \tau}{\sigma |B|} \mathbf{B} \times \mathbf{j} \tag{6.103}$$

There are two components of $\mathbf{E}$, one parallel to $\mathbf{j}$, i.e. $\mathbf{E}_x$ and the other perpendicular to $\mathbf{j}$, i.e. $\mathbf{E}_y$.

$$|E_x| = \frac{|j_x|}{\sigma} \tag{6.104}$$

$$|E_y| = R_H\, |\mathbf{B}|\, |\mathbf{j}_x| \tag{6.105}$$

with

$$R_H = \frac{\omega_c \tau}{\sigma B_z} = -\frac{1}{ne} = \frac{E_y}{j_x B_z} \tag{6.106}$$

where ω_c is the cyclotron frequency given by eB_z/m.

Relation (6.106) is the usual expression derived on the basis of much simple arguments. The experimental determination of R_H is based on this relation. The quantities E_y, J_x and B_z are all actually measured. The method of measurement follows from the geometry of Fig. 6.2, described earlier. Whereas B_z is obtained from the calibration curve of the electromagnet's field for the used pole spacing, E_y and J_x are derived from the measured voltage and current respectively along the y and x directions with the knowledge of sample dimensions. For a certain concentration of electrons, the calculated value of R_H ($= -\frac{1}{ne}$) as obtained from (6.106) is about 15 per cent lower than the value given by the Lorentz expression (6.45a), bringing it closer to the measured value. The flagrant discord for the transition metals, however, obscures this marginal success (see Table 6.3).

Furthermore, it would be too naive to take the accuracy of relations (6.103) and (6.106) for granted. In fact, the Hall coefficient and the resistivity both are found to depend on magnetic field and are not constant as indicated by these equations. In the Drude model one expects the Hall field (E_y) to be proportional to the magnetic field (B_z) and the current density (j_x). This keeps the Hall coefficient (R_H) constant. It is rather unbelievable that R_H should not depend on the relaxation time that strongly depends on temperature and the details of sample preparation. The carrier concentration as derived from the measured values of R_H matches well with the Drude's estimates only for alkali metals. Some metals show even positive Hall coefficient (see Table 6.3). All these questions including the magnetic field dependence are answered in the framework of band theory, which relies heavily on the fact that in real solids there are two types of constant energy surfaces—the electron and hole surfaces. In this model both the Hall coefficient and the resistivity and hence the magnetoresistance are shown to depend on the magnetic field and the relaxation time the details of which can be found in Chapter 9. The discussion at this stage, however, cannot be closed without stating that the Hall coefficient and the magnetoresistance do saturate at high magnetic fields under certain conditions to be specified in Chapter 9. An elaborate theory demonstrates that for many metals, this limiting value of Hall coefficient is precisely the same as given by Drude theory.

6.7 MATTHIESSEN'S RULE

The knowledge of different mechanisms of electron collisions is most crucial to the calculation of transport properties of solids. Electrons are mostly scattered by lattice phonons and electrons; and defects involving the impurities and the immobile crystal imperfections. The experimental behaviour of most of the metals reveals that the former processes dominate at room temperature and the latter at low temperatures. The two distinct mechanisms, as we identify, to a good approximation may be treated as independent. This allows us to write the net rate at which the momentum distribution would relax back to the equilibrium distribution after the electric field is switched off as

$$\frac{1}{\tau} = \frac{1}{\tau_l} + \frac{1}{\tau_i} \tag{6.107}$$

Using (6.101), we get

$$\frac{1}{\sigma} = \frac{1}{\sigma_l} + \frac{1}{\sigma_i}$$

or

$$\rho = \rho_l + \rho_i \tag{6.108}$$

where ρ denotes the resistivity with ρ_l and ρ_i referring to the contributions from the lattice and the impurities, respectively.

Therefore, in the presence of several scattering mechanisms the resistivity is simply the sum of the resistivities that we would have if each were present alone.

The conductivity plot of Fig. 6.1 and the relative resistance plots of Fig. 6.12 show a plateau below 10 K where the electron collisions with thermal phonons and electrons are almost frozen. The resistivity in this temperature range, that appears as temperature independent, is totally contributed by electron collisions with the impurities and the static imperfections. The value of ρ_i is estimated by extrapolating the curve back to 0 K in Fig. 6.12. The values of ρ_l at different temperatures are

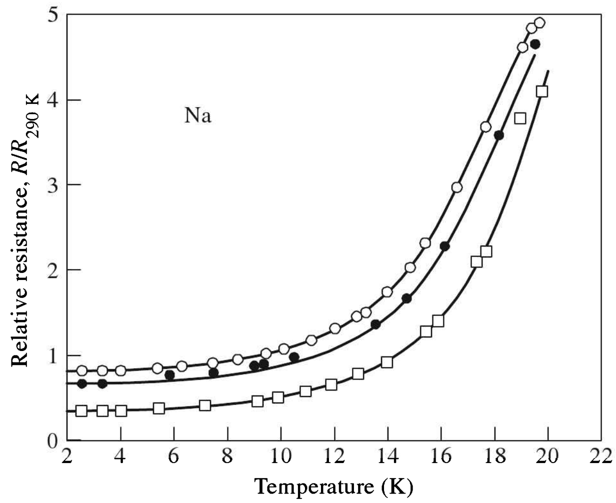

FIG. 6.12 Behaviour of electrical resistance of sodium compared to the value at 290 K with change in temperature. The experimental points marked as ○, ●, □ belong to three different samples with differing defect concentrations. [After D.K.C. McDonald, K. Mendelssohn, *Proc. R. Soc.*, Edinburgh Sect., A 202, 103 (1950).]

obtained by subtracting ρ_i from the measured values of ρ (6.108). The temperature dependence of ρ_l is in accordance with the theory of collisions of electrons with phonons and electrons. For small concentrations, ρ_l shows no dependence on the defect concentration.

The observation that ρ_i is temperature independent is established as the Matthiessen's rule. Matthiessen observed that the resistivity of a metal increases with the increase in defect concentration. This is confirmed by the experimental curves of sodium in Fig. 6.12 where the curve with higher intercept with the resistance axis refers to the sample of higher defect content.

The assumption that mechanisms of the two collision processes under consideration are independent is by and large vindicated by observations on whose basis the Matthiessen's rule is founded. But it is not difficult to show that the Matthiessen's rule breaks down even in the relaxation time approximation which depends on the electron wavevector **k**. On the other hand, the realistic picture of collisions warrants the assumptions of the relaxation time approximation to be set aside, as a result of which the two collision mechanisms have a rare chance of being independent and the Matthiessen's rule is in general rendered invalid. The presence of the two competing collision mechanisms can seriously affect the configuration of electrons that control the actual collision rate. It could only be fortuitous that the distribution function in the presence of each separate mechanism be the same and therefore, the failure of the Matthiessen's rule be prevented.

EXAMPLE 6.3 Assuming that the lattice vibrations of a crystal can be determined in the Einstein model, calculate the contribution from phonon scattering to the electrical resistivity.

Solution: In Einstein model all atoms vibrate independently with a common frequency ω_E and the energy exchanged with electrons is guided by

$$\hbar\omega_E = k_B\theta_E$$

The equation of motion of an atom in a given direction is written as:

$$M\ddot{s} = -fs, \text{ solving which one gets } \omega_E^2 = \frac{f}{M}$$

The mean energy in the each mode is given by:

$$<\varepsilon(\omega_E)> = f<s^2> = M\omega_E^2 <s^2>$$

Hence, the mean square displacement $<s^2> = \dfrac{<\varepsilon(\omega_E)>}{M\omega_E^2}$

$$= \frac{1}{M\omega_E^2} \frac{\hbar\omega_E}{\left(e^{\frac{\hbar\omega_E}{k_B T}} - 1\right)}$$

(in view of relations (5.15) and (5.16))

At high temperatures ($T > \theta_E$),

$$<s^2> = \frac{\hbar}{M\omega_E} \cdot \frac{k_B T}{\hbar\omega_E} = \frac{k_B T}{M\omega_E^2} = \frac{\hbar^2}{Mk_B\theta_E^2} \cdot T$$

Thus, the expected phonon contribution to the resistivity is linear with temperature and

$$\rho_{\text{phonon}}(T) \sim \frac{T}{M\theta_E^2}$$

This temperature dependence is confirmed by experiments. However, the Einstein model is unable to account for the observed temperature dependence at low temperatures ($T < \theta_E$).

6.8 THERMOELECTRIC EFFECTS

The thermoelectric behaviour of metals is described mainly in terms of three parameters: (i) the thermoelectric power, (ii) the Thomson coefficient, and (iii) the Peltier coefficient. These parameters will be defined in this section on the basis of the thermoelectric response of a conducting rod whose ends are maintained at different temperatures T_1 and T_2 ($T_1 < T_2$), as shown in Fig. 6.13.

6.8.1 Thermoelectric Power

Electrons at the hot end on the average are at higher energies. So, there are more electrons in levels above the Fermi level ε_F at the hot end. These electrons lower their energy by drifting to the cold end.† The hot end thus becomes positively charged and the cold end negatively charged, creating different electric potentials at the ends. The magnitude of the induced difference of potential (V_2) depends entirely on the temperature difference ($T_2 - T_1$) and remains unaffected by

† Even in metals a part of heat is conducted by phonons, which are the main carriers of heat in insulators [see Section 5.4.3].

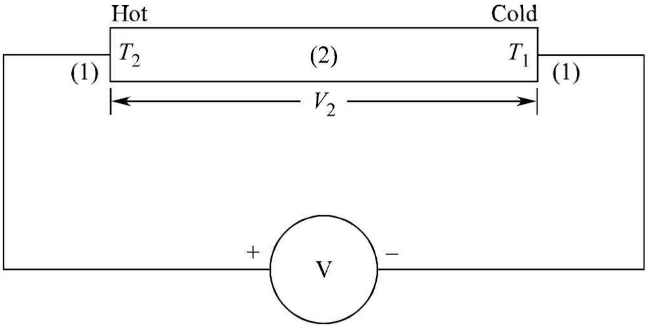

FIG. 6.13 Measurement of the induced voltage (thermo e.m.f.) in a conducting rod whose ends are maintained at temperatures T_1 and T_2 ($T_2 > T_1$).

any change in the temperature distribution along the rod. A voltmeter connected to measure the induced voltage shows no deflection if the connection wires (1) are of the same material as that of the rod. If this material is different, the induced potential difference between the ends of the wires (1), in contact with the ends of the rod, is no more V_2 but some other value, say V_1. The voltmeter reading in this case will be $(V_1 - V_2)$, where V_1 and V_2 refer to the difference of potentials in open circuit.

The difference of potential $(V_1 - V_2)$ denoted by V_{12}, rises monotonically with the temperature difference $(T_2 - T_1)$. The rate of change of the induced voltage with the temperature difference defines the thermoelectric power S_{12} of the junction (1, 2). If ΔV_{12} be the rise in V_{12} because of the small increase ΔT in $(T_2 - T_1)$, then

$$S_{12} = \frac{dV_{12}}{dT} = \frac{dV_1}{dT} - \frac{dV_2}{dT} = S_1 - S_2 \tag{6.109}$$

where S_1 and S_2 are the bulk thermoelectric powers of materials to which the wires and the rod, respectively, belong. The V_{12} is often referred to as the ***Seebeck potential.*** Relation (6.109) most significantly emphasizes that the thermoelectric power of a junction is not a property of the junction. It is immaterial whether the junction is soldered, brazed, spot-welded or fused; S_{12} depends only on the bulk properties S_1 and S_2 of the materials in contact at the junction.

6.8.2 The Thomson Effect

Let an electric current be forced into the rod (Fig. 6.14) between its ends. When the conventional current flows down the rod from the hot end to the cold end, the electrons reaching the hot end raise their potential energy by absorbing heat from the hot end. Similarly, for the reverse conventional current, the electrons lower their potential energy at the cold end by emitting heat as

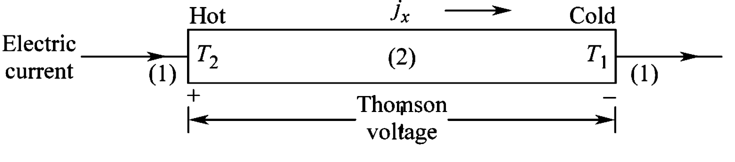

FIG. 6.14 Schematic of the Thomson effect. The Thomson voltage produced because of the temperature gradient is in addition to the Ohm's law voltage.

they arrive there from the hot end. Thus there occurs an absorption or evolution of heat on the passage of current through a rod whose ends are maintained at different temperatures. This amount of heat is in addition to the power loss caused by Joule's heating and the heat flowing because of the presence of a temperature gradient. This absorption or evolution of heat is known as the ***Thomson effect.*** It does not depend on the nature of the junction involved.

It is observed that the heat evolved per unit volume per unit time dQ/dt is proportional to the temperature gradient $-\,dT/dx$ for a constant current density j_x down the length of the rod (see Fig. 6.14). Also, under the condition of a fixed temperature gradient dQ/dt is found to vary linearly with j_x. These variations may accordingly be expressed as

$$\frac{dQ}{dt} \propto -\frac{dT}{dx}, \text{ with current density } j_x \text{ maintained at a constant value.}$$

$$\frac{dQ}{dt} \propto j_x, \text{ with } -\frac{dT}{dx} \text{ maintained at a fixed value.}$$

Therefore, we may express dQ/dt as

$$\frac{dQ}{dt} = -\mu_T j_x \left(\frac{dT}{dx}\right) \tag{6.110}$$

where the constant of proportionality μ_T is called the ***Thomson coefficient.***

The sign of Q, the heat evolved, is taken as positive here.

6.8.3 The Peltier Effect

An electric current density j_x through a conductor down its length (in the x-direction) at constant temperature always has a heat current density j_Q associated with it such that

$$j_Q \propto j_x$$

or

$$j_Q = \Pi j_x \tag{6.111}$$

where Π is called the *Peltier coefficient.*

Relation (6.111) implies that the heat current density will undergo a change at a junction of two conductors 1 and 2 when an electric current flows across the junction because Π is different for the two conductors. The change in heat flux results in absorption or evolution of heat at the junction, depending on the direction of the electric current. When an electric current is forced into a thermocouple, the heat is absorbed at one junction and evolved at the other because the direction of the electric current between conductor 1 and conductor 2 is reversed at the other junction (Fig. 6.15). This is known as the *Peltier effect.* The Peltier effect essentially involves the pumping of heat energy from one junction to the other by making an electric current flow through a thermocouple.

It is appropriate to denote the Peltier coefficient at a junction by Π_{12} and define it as the reversible heat evolved or absorbed per unit time per unit electric current flowing from conductor 1 to conductor 2 at their junction. According to this definition, we have

$$\Pi_{12} = -\,\Pi_{21} \qquad \text{with} \qquad \Pi_{12} = \Pi_1 - \Pi_2$$

Like the Thomson effect, neither the Peltier effect nor the Seebeck effect depends on the nature of the junction.

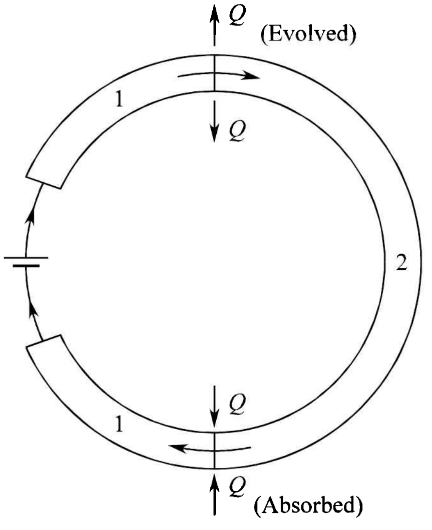

FIG. 6.15 Schematic of the Peltier effect. The heat is shown to be evolved at a thermocouple junction 1–2 when a current flows from the conductor 1 to the conductor 2. The heat is absorbed at the junction 2–1 when the current flows from the conductor 2 to the conductor 1 (the current direction is reversed).

6.8.4 Kelvin (Thomson) Relations

Thomson (famous as Lord Kelvin) used the principles of classical thermodynamics to show that the three thermoelectric parameters S, μ_T and Π are not independent and that only one is needed to specify the other two. He derived the following relations, commonly known as Kelvin relations:

$$\mu_T = T\frac{dS}{dT} \tag{6.112}$$

$$\Pi = TS \tag{6.113}$$

In the above relations, S stands for the absolute thermoelectric power and not the entropy. Although the laws of classical thermodynamics are not directly applicable to thermoelectricity, the Kelvin relations stand the test of experiments extremely well. If either S or μ_T is known as a function of temperature, the remaining two parameters can be obtained from (6.112) and (6.113). In the latter case, relation (6.112) is integrated to get the absolute value of S at any temperature, i.e.

$$S(T) = \int_0^T \frac{\mu_T}{T}\, dT \tag{6.114}$$

Here, we have ignored $S(0)$ because the third law of thermodynamics implies that all thermoelectric effects vanish at absolute zero.

A few striking gains notwithstanding, the limitations of the free electron theory are pronounced by its varying degree of success in accounting for several properties of metals. Some of these might not have even figured in this chapter. Satisfactory explanations to all such properties and those of solids in general are provided by a less naive theory which forms the subject of the next two chapters.

SUMMARY

1. According to the Drude model, the d.c. electrical conductivity of metals is

   $\sigma = \dfrac{ne^2\Lambda}{(3m\,k_B T)^{1/2}}$, where Λ is the electron mean free path.

   Experimentally, it is observed that

$$\sigma \propto T^{-1}.$$

2. The Drude model correctly describes the Wiedemann–Franz law, but gives a wrong magnitude of the Lorenz number L. The value of L as defined in the Drude theory is about half the measured value and is given by

   $L = \dfrac{K}{\sigma T} = \dfrac{3}{2}\left(\dfrac{k_B}{e}\right)^2$, where K is the thermal conductivity.

3. The dc electrical conductivities in the Drude model and the Lorentz model are related as

   $\sigma_D = \left(\dfrac{3\pi}{8}\right)^{1/2}\sigma_L$, D: Drude

   L: Lorentz

4. According to the Lorentz model, the Hall coefficient of metals is given by

$$R_H = -\left(\frac{3\pi}{8}\right)\frac{1}{ne}$$

   The measured values of the Hall coefficient are close to the values given by

$$R_H = -\frac{1}{ne}$$

   R_H is measured using the formula:

   $R_H = \dfrac{E_y}{j_x B_z}$, where the Hall field E_y, the current density j_x and the magnetic field B_z are mutually perpendicular along the y, x and z directions, respectively.

5. The Fermi–Dirac distribution function is

$$f(\varepsilon) = \frac{1}{\exp\left[(\varepsilon - \mu)/k_B T)\right] + 1}$$

   where μ is the *chemical potential,* defined by $\lim\limits_{T \to 0} \mu = \varepsilon_F$; ε_F is the Fermi energy.

6. In the Sommerfeld model for metals:

(i) The density of states is

$$D(\varepsilon) = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\varepsilon^{1/2}\left(= \frac{3N}{2\varepsilon_F} \text{ at } \varepsilon = \varepsilon_F\right)$$

with $\varepsilon_F = \dfrac{\hbar^2}{2m}(3\pi^2 n)^{2/3}$, where $n = \dfrac{N}{V}$.

(ii) The electron heat capacity is

$$C_{el} = \frac{\pi^2}{2} nk_B\left(\frac{T}{T_F}\right)$$

(iii) The dc electrical conductivity is

$$\sigma_0 = \frac{ne^2\tau(\varepsilon_F)}{m} = \frac{ne^2\Lambda_F}{mv_F}$$

and the ac conductivity is,

$$\sigma = \frac{\sigma_0}{(1 - i\omega\tau)}$$

7. The resistivity of a metal is expressed as

$$\rho = \rho_l + \rho_i$$

where l and i distinguish the lattice (or phonon) and impurity contributions.

The observation that ρ_i is temperature independent is established as the Matthiessen's rule.

8. Heat absorbed or evolved in the Thomson effect is in addition to the energy loss caused by Joule's heating (produced owing to the electric current, forced externally) and the heat flowing by virtue of the temperature gradient.

9. The commonly known Kelvin relations are:

$$\mu_T = T\frac{dS}{dT}$$

$$\Pi = TS$$

where μ_T is the Thomson coefficient and S and Π denote the thermoelectric power and the Peltier coefficient, respectively.

PROBLEMS

6.1 The mean free time for electrons in a trivalent metal is 5×10^{-14} s. Its atomic weight is 27 and density 2.7×10^3 kg m^{-3}. Find out the current flowing through its wire 9 m long and of 0.5 mm^2 cross-sectional area when a difference of potential of 1.4 V is impressed across its ends.

6.2 A current of 3 A flows along the length of a metal bar with rectangular cross-section of 6 mm^2. When a static magnetic field of $2T$ is applied along one side of its cross-sectional face, a Hall voltage of 10 μV develops between a pair of parallel surfaces that are separated by the other side 2 mm long. Calculate the Hall coefficient and the current carrier concentration.

6.3 An electron in a cubic three-dimensional box falls from the second excited state to the ground state. Calculate the amount of energy emitted in the process. Assume that the energy appears as radiation, determine the corresponding wavelength. Take the box side as 0.3 mm long.

6.4 Assuming Drude model:
(a) Show that the probability for an electron, picked at random at any time, not to have made a collision during the preceding t is given by e^{-t}/τ.
(b) Show that at any moment, the time between the last and the next collision averaged over all electrons is 2τ. In view of this result, comment on the Drude's mistake of a factor of 1/2 in the calculation of electrical conductivity.

6.5 Consider a spherical body of mass m and charge q moving through a viscous fluid with a constant velocity ($\propto q\mathbf{E}_x$) under the action of an electric field $\mathbf{E}_x$. A small magnetic field $\mathbf{B}_z$ is now applied. Obtain equations of motion in the xy-plane and find the radius of curvature of the trajectory.

6.6 (a) Plot $f(\varepsilon)$ as a function of $(\varepsilon - \varepsilon_F)$ in units of k_BT at room temperature in the range $-3k_BT \leq \varepsilon \leq 3k_BT$.
(b) Examine at what values of ε, the $f(\varepsilon)$ differs appreciably from unity.

6.7 Give a comparison of the equilibrium distribution functions representing Maxwell–Boltzmann and Fermi–Dirac statistics. Explain why only the former depends on the particle density.

6.8 Calculate the density of states for a two-dimensional gas of free electrons in a so-called quantum well. The boundary conditions for the electronic wave function are: $\psi(x, y, z) = 0$ for $|x| > a$, where a is of atomic dimensions.

6.9 Show that the fraction of electrons within k_BT of the Fermi level is equal to $3k_BT/2\varepsilon_F$, if $D(\varepsilon) \sim \varepsilon^{1/2}$.

6.10 The magnitude of the electron wavevector corresponding to the top most level filled in a system of free electrons is equal to $(2\pi^2)^{1/3}/a$, where a is the interatomic separation. If each level is three-fold degenerate, calculate the number of conduction electrons per atom.

6.11 Derive relations for the electron density at which the Fermi surface first touches the zone boundary in Na and Cu metals.

6.12 Use the free electron theory to calculate the Fermi energy of Na and Al metals. Their lattice constants are 4.3 Å and 4.0 Å, respectively.

6.13 Calculate the Fermi energy and Fermi temperature of liquid He^3 whose density near 0 K is 81 kg m^{-3}. (He^3 is a fermion with atomic spin 1/2.)

6.14 At what temperature does the electron heat capacity become larger than the phonon heat capacity? Express this temperature in terms of the Debye temperature and the electron density.

6.15 Calculate the electron heat capacity at 1000 K in Na, Al and Cu metals using the following data:

	Electron density $n(\times 10^{28}\ m^{-3})$	*Fermi energy* ε_F(eV)
Na	2.5	3.1
Al	18.0	11.7
Cu	8.5	7.1

Estimate the fraction it forms of the total heat capacity, assuming that 1000 K is well above the Debye temperature in each case.

6.16 The electrical conductivity of a pure gold crystal is 5×10^7 ohm^{-1} m^{-1} at 273 K. Gold has a close-packed FCC structure with atomic radius as 1.44 Å. Each atom contributes one conduction electron. If a gold crystal has 0.1 per cent of randomly distributed vacancies, estimate the electron mean-free path as determined by scattering from vacancies alone. Calculate the mean-free path for phonon-scattering at 273 K from the given value of conductivity. What is the revised value of conductivity according to Matthiessen's rule?

SUGGESTED FURTHER READING

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